

Universal properties from quantum many-body dynamics

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The Loschmidt echo—the amplitude for a quantum state to return to itself—provides a dynamical route to obtain universal properties of a critical many-body system. Analytically continuing from real to imaginary time, the echo becomes the partition function of a conformal field theory on a strip. The central charge and operator content of the corresponding conformal-field theory are then encoded in the spectrum of a transfer matrix, obtained by contracting the space–time tensor network along the space direction. The cost of this transverse contraction is governed by the temporal entanglement, which grows only logarithmically at criticality, allowing the method to reach times where conventional evolution algorithms are limited by the entanglement barrier. These results were established for integrable chains, and this thesis extends them to the non-integrable regime. We study a transverse-field Ising model extended by next-nearest-neighbour interactions, which remains critical and in the Ising universality class at weak frustration. We encode its evolution as a matrix product operator, and we resolve the leading spectrum of the resulting non-Hermitian transfer matrix with a block power method that evolves its left and right eigenvectors together. Where the method allows, we recover the conformal signatures: the central charge remains consistent with the Ising value $c = \frac{1}{2}$, and the boundary operator content of the Ising class is reproduced. Throughout, we separate the universal quantities from the non-universal lattice constants that accompany them, and we map where and why the method reaches its limits. *This is a working draft.*

Keywords: Tensor Networks, Transverse Contraction, Many-body Systems, Quantum Dynamics, CFT

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I thank my dad, my brother, but specially to my mom, my gf Helena, my flatmates...

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Acronyms

CFT	Conformal Field Theory	2
DMRG	Density Matrix Renormalization Group	1
ETH	Eigenstate Thermalization Hypothesis	1
FSM	finite-state machine	III
JW	Jordan–Wigner	10
MPO	Matrix Product Operator	6
MPS	Matrix Product State	1
NN	nearest-neighbour	12
NNN	next-nearest-neighbour	2
RDM	Reduced Density Matrix	8
RTM	Reduced Transition Matrix	III
SVD	Singular Value Decomposition	28
TDVP	Time-Dependent Variational Principle	1
TEBD	Time-Evolving Block Decimation	1
TN	Tensor Network	1
tMPO	temporal Matrix Product Operator	6
tMPS	temporal Matrix Product State	14

1 Introduction and Motivation

Following a closed quantum system after being driven far from equilibrium is one of the central questions of modern many-body physics. Simulating such dynamics is difficult, but it can also be revealing: at criticality, the real-time evolution of a simple initial state can expose universal information normally associated with low-energy physics. Our aim with this thesis is to determine how far this dynamical route can be pushed once the system is no longer exactly solvable.

This question is particularly relevant for non-integrable systems, which lack exact analytical solutions. Their dynamics is also intrinsically harder to simulate over long times: after a quench these systems are generically expected to thermalize, acting as their own heat bath so that local observables relax to values fixed only by the energy—the content of the Eigenstate Thermalization Hypothesis (ETH) [1–3]—whereas integrable models, protected by extensively many conserved charges, retain memory of their initial state instead [4]. Analytical results are largely confined to those integrable limits, so almost everything we know about thermalization, scrambling of the quantum information and the onset of chaos in non-integrable models depends on our ability to simulate their dynamics numerically.

Exact numerical simulations are severely limited by the exponential scaling of the Hilbert space, which makes representing a generic many-body wavefunction intractable. To circumvent this bottleneck, the community has turned to Tensor Networks (TNs), a framework that efficiently compresses quantum states by using entanglement as the organizing principle. For one-dimensional systems in equilibrium, Matrix Product States (MPSs) combined with algorithms such as the Density Matrix Renormalization Group (DMRG) have become the state-of-the-art approach [5–7]. Their efficiency comes from the area law of entanglement, which holds for ground states of one-dimensional systems with local gapped Hamiltonians: the entanglement entropy across a bipartition saturates to a constant rather than growing with the subsystem size [8]. Since the required MPS bond dimension is controlled by this entanglement, the ansatz efficiently targets the low-entanglement corner of Hilbert space where these ground states lie.

Simulating dynamics, however, runs into the so-called entanglement barrier. After a quantum quench, the entanglement entropy typically grows linearly in time, approaching volume-law behaviour [9]. Although highly optimized algorithms such as Time-Evolving Block Decimation (TEBD) [10] and the Time-Dependent Variational Principle (TDVP) [11] are very effective at short times, faithfully representing this growth requires a bond dimension that increases exponentially with time, making standard simulations exponentially costly at late times [12, 13].

To overcome this barrier, transverse contraction is a framework that adopts a different perspective [14, 15]. Rather than evolving a state forward in the Schrödinger picture, it represents the full dynamics as a two-dimensional tensor network and contracts it along the original spatial direction. The computational cost is then controlled by the temporal entanglement entropy, defined through a Schmidt decomposition across a cut in time rather than in space. Although its physical interpretation remains an open question, it is computationally well defined. For a generic quench it may grow as strongly as spatial entanglement; nevertheless, the transverse viewpoint can provide a useful way to tackle specific dynamical quantities.

The central quantity we compute throughout is the Loschmidt echo for an infinite, translation-invariant system: the amplitude for a state to return to itself after evolving for a time T [16]. It measures how quickly the evolved state departs from, and how coherently

it returns towards, its initial condition, providing a natural measure of relaxation after a quench. Given its structure, transverse contraction offers an efficient way to compute it by contracting the space–time network directly with a power method [17].

At a conformally invariant critical point, the echo also provides direct access to universal data such as the central charge of the underlying Conformal Field Theory (CFT). In the transverse picture, it is governed by a spatial transfer matrix whose spectrum is predicted by the CFT and which becomes effectively unitary at long times—a phenomenon known as emergent dual unitarity [18]. This behaviour has been demonstrated explicitly for the critical Ising and three-state Potts chains [17], together with the finite-time corrections [19] that we use in Section 7.

The central question is therefore whether these conformal signatures survive the loss of integrability of the model. To address it, we study a self-dual quantum spin chain in the thermodynamic limit that extends the transverse-field Ising model with next-nearest-neighbour (NNN) interactions controlled by a coupling p [20]. For any $p > 0$ it is non-integrable and longer-ranged, making its dynamics costly for conventional methods, yet it remains critical and in the Ising universality class over a wide range of p [20], so the CFT predictions provide a clear target for numerical simulations. We prepare the system in the fully polarized paramagnet state, quench it to the critical point, and use the transverse contraction machinery to compute the Loschmidt echo and the generalized temporal entropies, which we compare with the CFT predictions.

The rest of this thesis is organized as follows. Section 2 defines the Loschmidt echo and develops the CFT framework that relates the transfer-matrix spectrum and generalized temporal entropy to the central charge and operator content. Section 3 develops the transverse-contraction framework: the rotated picture, the temporal states, and the transition matrix from which the generalized temporal entropies are built. Section 4 introduces the ANNNI-type model, establishes its critical Ising universality, and measures the frustration-dependent sound velocity. Section 5 constructs the time-evolution MPO and its non-Hermitian transfer matrix. Section 6 presents the block power method, and Section 7 validates the numerical results with the CFT predictions.

2 Universal Data from the Dynamics

The real-time dynamics of a critical lattice model contains universal information that can be extracted from a simulation. At a continuous critical point, the lattice model forgets its microscopic details and is described by a CFT—a continuum theory so constrained by symmetry that its predictions are fixed by a small set of numbers: the central charge c and the spectrum of scaling dimensions of its operators. These quantities are universal: every model in the same universality class reproduces them, which is exactly what makes them good targets for a numerical method. This section relates the Loschmidt echo to a boundary CFT, states the resulting universal predictions, and identifies the non-universal lattice parameters needed to test them.

2.1 From the Loschmidt echo to a boundary CFT

The object we compute is the Loschmidt echo,

$$\mathcal{L}(T) = \langle \Psi_0 | e^{-iHT} | \Psi_0 \rangle. \quad (1)$$

For an infinite, translation-invariant chain prepared in $|\Psi_0\rangle$, this is the amplitude to return to the initial state after evolving for a time T with the critical Hamiltonian. We take $|\Psi_0\rangle$

to be a product state; its particular choice determines the boundary condition discussed below.

To connect this amplitude with the continuum theory, we rotate the evolution time to the imaginary axis, $T \rightarrow -i\tau$ (a Wick rotation [21]). The real-time propagator e^{-iHT} then becomes the Euclidean weight $e^{-\tau H}$, the same operator that generates the thermal partition function of a two-dimensional statistical system. The matrix element in Eq. (1) can consequently be written as a Euclidean path integral on a slab of width τ , with both edges fixed by $|\Psi_0\rangle$. A state that can terminate a path integral in this way, without introducing any scale of its own, is what a **CFT** calls a conformal boundary state. The echo is therefore the partition function of the critical theory on a long strip whose two edges carry the boundary condition set by $|\Psi_0\rangle$ and whose width is the (imaginary) evolution time. The rotation is used only to obtain the conformal predictions, which are derived on the Euclidean strip; they are continued back to real time before being compared with the simulation. Figure 1 shows this construction, first as a tensor network and then as the strip it represents.

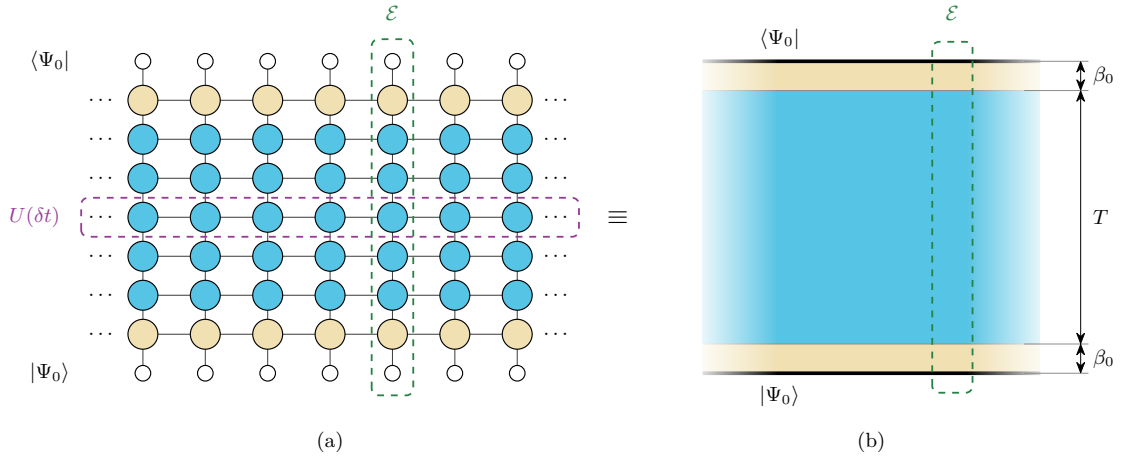


Figure 1: **The Loschmidt echo as a tensor network and as a conformal strip.** (a) Trotterising the evolution expresses the echo as the contraction of a two-dimensional network. Open circles are the product state $|\Psi_0\rangle$ closing the network at either end; each row of filled circles is one Trotter layer $U(\delta t)$, and the cream rows at top and bottom represent the imaginary-time cooling β_0 that brings the bare product state towards a conformal boundary condition. (b) The same object after the Wick rotation, in the continuum. The heavy lines are the edges where $|\Psi_0\rangle$ closes the path integral, the cream bands are the cooling intervals and the blue bulk is the real-time evolution, so the strip spans $T + 2\beta_0$ in the time direction and is infinite in the spatial one.

Two features of this construction are important for the numerics. First, a bare product state carries weight on high-energy configurations and is not itself a conformal boundary state. We therefore cool it in imaginary time for a short interval β_0 , which suppresses this high-energy content and brings the state into the regime where the **CFT** applies. On the lattice, the cooling contributes $N_\beta = 2\beta_0/\delta t$ imaginary-time sites split between the two ends, so the strip spans $T + 2\beta_0$ in the time direction. Second, the initial state is the boundary condition of the strip: each $|\Psi_0\rangle$ realizes a particular conformal boundary condition, whose lightest boundary operator has a dimension x_1 that controls how quickly finite-size corrections decay. For instance, in the Ising class, the disordered (free) boundary has $x_1 = \frac{1}{2}$ and converges quickly, whereas the ordered (fixed) boundary has $x_1 = 2$ and converges slowly. In the σ^x -field convention, the polarized $|X+\rangle$ initial state realizes the free boundary, which is why we quench from it.

We contract the strip in the transverse direction. In the Euclidean description, space and imaginary time are coordinates of the same two-dimensional geometry, so either one can be chosen as the direction of propagation. In the transverse picture, we exchange their usual roles and contract the network along the original spatial direction. This defines a transfer matrix $\mathcal{E}$ that propagates from one spatial column to the next while acting on states supported on a temporal slice. Its tensor-network construction is developed in Section 3.

In two dimensions, conformal symmetry strongly constrains both the spectrum of $\mathcal{E}$ and the entanglement across a cut in the strip [22, 23]. These two observables provide independent routes to the universal data; Appendix A.1 summarises the conformal mappings used to derive them.

2.2 The universal predictions

Confining a CFT to an infinite strip of transverse size β shifts its ground-state energy by an amount set by the central charge. The excited levels above this shift are organised by the operator scaling dimensions x_i . The transfer matrix inherits this structure [17],

$$\mathcal{E} = e^{-a\hat{H}} = \exp\left[-(-\kappa + \pi L_0)g\right], \quad \kappa = \frac{\pi c}{24}, \quad (2)$$

where L_0 is the operator whose eigenvalues list the scaling dimensions, κ is the Casimir number, and $g = a/\beta$ is the ratio of the column spacing to the strip width, with $a = 1$ in lattice units. The width is complex, $\beta = v(2\beta_0 + iT)$, because the regulator contributes imaginary time while the quench contributes real time. Substituting this width and expanding for $\beta_0 \ll T$ shows that the universal contributions enter the eigenvalue phases at order $1/T$ but affect their moduli only at order β_0/T^2 (Appendix A.2). A single diagonalisation can therefore determine both c and the dimensions x_i from the phases.

The second route uses entanglement. In a CFT, the Rényi entropies follow from correlators of twist operators whose scaling dimension is proportional to c [24]. In the transverse setting, the cut runs across the strip in the time direction, and its two sides do not belong to a single quantum state. The entropies are instead built from the left and right temporal MPSs of Section 3; they are therefore generalized entropies [25] and are generally complex. For a cut at time t [17, 19],

$$S_n^{\text{gen}}(t) = s_0 + \frac{i\pi c}{24}\left(1 + \frac{1}{n}\right) + \frac{c}{12}\left(1 + \frac{1}{n}\right)W(t, T), \quad (3)$$

with the chord variable

$$W(x, \ell) = \log\left[\frac{2\ell}{\pi} \sin \frac{\pi x}{\ell}\right], \quad (4)$$

where x is the position of the cut and ℓ is the extent of the system. Evaluating W on the interval $(0, T)$ makes the evolution time play the role of the subsystem size in the spatial formula. The entropy therefore grows as $\log T$, and the bond dimension required for the contraction grows only polynomially with T [12, 13]. We use the Rényi-2 member throughout because it is the least expensive to compute and requires only two copies of the system, making it experimentally accessible [26]. The equilibrium calculations of Section 4 instead use the ground-state von Neumann entropy, which DMRG provides directly from the entanglement spectrum.

Equations (2) and (3) give four predictions, all tested in Section 7. Three concern the transfer eigenvalues μ_i . We work with their logarithms, $\lambda_i = \log(-\mu_i)$, so that $\text{Im } \lambda_i = \arg(-\mu_i)$ is the phase of μ_i measured from the negative real axis. The minus sign shifts

every phase by the same constant π ; this shift cancels in phase differences and is absorbed by a constant when fitting $\text{Im } \lambda_0$ alone.

The generalized-entropy profile. Separating Eq. (3) into real and imaginary parts gives

$$\text{Re } S_n^{\text{gen}}(t) = s_0 + \frac{c}{12} \left(1 + \frac{1}{n}\right) W(t, T), \quad \text{Im } S_n^{\text{gen}} = \frac{\pi c}{24} \left(1 + \frac{1}{n}\right). \quad (5)$$

The real part follows a symmetric chord profile. Plotted against $W(t, T)$, it becomes a straight line whose known slope determines c (Section B). The imaginary part is independent of t and gives a second estimate of c without fitting the spatial profile. Finite-time corrections to S_n^{gen} decay as $T^{-1/n}$ [19]; for the Rényi-2 entropy used here this is $T^{-1/2}$, so extracting c requires extrapolation to the long-time limit.

The central charge from the leading eigenvalue. The central-charge and boundary-gap relations are expansions in $1/T$ of the logarithms λ_i , rather than of the eigenvalues themselves. Substituting the strip width into Eq. (2) and expanding gives the following relation [17], derived in Appendix A.2:

$$\frac{\text{Im } \lambda_0}{T} = a_0 + \frac{C}{T^2} + \frac{a_4}{T^4} + \dots, \quad C = -\frac{\kappa}{v} = -\frac{\pi c}{24 v}. \quad (6)$$

Here a_0 and a_4 are non-universal coefficients, while the coefficient of $1/T^2$ carries the Casimir number divided by the velocity. Once v has been measured independently (Section 4.2), the relation $c = 24 v |C|/\pi$ determines the central charge from the eigenvalue phase alone, without relying on the entropy. Because this term is small, the extraction requires a nonzero β_0 and sufficiently long times. In practice we fit $\text{Im } \lambda_0 = AT + B + C/T + D/T^3$ on the unwrapped phase, where B absorbs the branch shift.

The boundary exponent from the first gap. The gap between the leading and first subleading eigenvalues measures the lightest boundary dimension,

$$\text{Im}(\lambda_1 - \lambda_0) = \frac{\pi x_1}{v T}. \quad (7)$$

Once the sound velocity is known, this phase gap determines x_1 . More generally, each subleading eigenvalue corresponds to a boundary dimension x_i through the same relation. We refer to this spectrum—the dominant eigenvalue followed by one subleading eigenvalue for each boundary operator, ordered by increasing phase distance—as the conformal tower, and to its members as tower states.

Emergent dual unitarity. Equation (2) also fixes how the spectrum evolves with T . Unlike the two predictions above, this one is an expansion of the eigenvalues themselves rather than of their logarithms: the moduli of the leading eigenvalues approach a common value while their phases stay distinct,

$$\frac{|\mu_i(T)|}{|\mu_0(T)|} = 1 - \frac{a_i}{T} - \frac{b_i}{T^2} - \dots \xrightarrow{T \rightarrow \infty} 1. \quad (8)$$

Equal moduli with distinct phases indicate that the temporal transfer matrix approaches a unitary operator up to an overall non-universal normalisation. We test this behaviour by following the dominant eigenvalue $\mu_0(T)$ in the complex plane as it moves along a trajectory of nearly constant radius with a winding phase.

These predictions apply to the continuum theory, whereas a simulation contains additional lattice scales and numerical parameters. Section 3 constructs the transfer and transition matrices used to measure the predicted quantities, and Appendix B explains how the universal contributions are separated from the non-universal parameters before the comparison in Section 7.

3 Time Evolution via Transverse Contraction

Section 2 showed that the Loschmidt echo of a critical quench encodes conformal data in the spectrum of a transfer matrix acting along the spatial direction of the corresponding strip. We now construct this transfer matrix and explain how it is used numerically. Transverse contraction was introduced for infinite chains by representing the dynamics with MPS extending along the time direction [14, 15], and was subsequently developed to study temporal entropies [17, 27]. Our calculations use the `ITransverse.jl` implementation developed in [28].

We rotate the space-time tensor network and contract it along the spatial direction. This reformulation maps the contraction to an eigenproblem for the spatial transfer matrix, whose dominant eigenvectors define the temporal boundary states. These states determine the RTM used to compute the generalized temporal entropies. We finally compare two truncation schemes for controlling their bond dimension.

3.1 The rotated network

In the transverse contraction framework, the time evolution of a D -dimensional system is represented by a $(D+1)$ -dimensional tensor network, with the additional dimension corresponding to time. For the one-dimensional chain studied here, the result is a two-dimensional network with no open physical legs. The echo is a scalar amplitude, so this network is analogous to the partition function of a two-dimensional classical model. Its exact contraction is #P-hard in general [29], which requires us to use an approximate representation with a controlled bond dimension (Section 3.4).

The tensor-network representation of the echo follows directly from Eq. (1) and is pictured in Figure 1a. The initial and final states are represented as MPSs, while the evolution operator is decomposed into a sequence of Matrix Product Operator (MPO) layers, one for each Trotter step. Because we require the Hamiltonian to be translation invariant and time independent, the same bulk tensor W appears at every spatial site and time step. Section 5 constructs this tensor for the model with NNN interactions.

In conventional evolution algorithms, successive MPO layers of the time-evolution operator $U(\delta t)$ are applied to the initial MPS $|\Psi_0\rangle$. The resulting spatial entanglement grows linearly with time (volume law) and eventually makes the state intractable at fixed bond dimension [9, 12, 13]. Transverse contraction instead contracts the same network column by column along the spatial direction. The tensors remain unchanged, but their indices exchange roles: the virtual spatial bonds become physical temporal indices, while the physical spatial indices become virtual temporal bonds. Figure 2 illustrates this exchange. The bond dimension of $U(\delta t)$ therefore sets the physical dimension of the temporal problem. Each bulk column defines a temporal Matrix Product Operator (tMPO) $\mathcal{E}$ that acts as a transfer matrix along the original spatial direction, and translation invariance ensures that the same $\mathcal{E}$ is applied at every site.

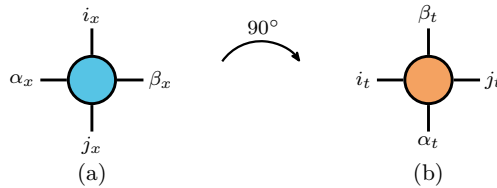


Figure 2: **The rotation exchanges physical and virtual indices.** A rank-4 tensor read (a) along space, with (i_x, j_x) physical and (α_x, β_x) virtual, and (b) along time, where the roles are exchanged.

These objects provide the lattice representation of the strip geometry introduced in Section 2. The strip is infinite along the spatial direction, and applying infinitely many columns on either side of a cut projects the states $\langle L|$ and $|R\rangle$ onto the dominant left and right eigenvectors of $\mathcal{E}$. The resulting fixed points are independent of how the chain is terminated far from the cut. They encode the temporal boundary condition imposed by $|\Psi_0\rangle$ at the top and bottom edges. The strip width is set by the number of temporal sites, including the real-time evolution and the cooling sites associated with the regulator β_0 . The CFT constrains the spectrum of the column transfer matrix $\mathcal{E}$, while the overlap $\langle L|R\rangle$ gives the Loschmidt echo after the spatial contraction.

The row and column transfer operators have different spectral properties. Each row represents the unitary Trotter step $U(\delta t) = e^{-iH\delta t}$ and therefore has eigenvalues of unit modulus. By contrast, the column transfer matrix $\mathcal{E}$ is non-Hermitian and generally has complex eigenvalues with different moduli. The implications of its non-Hermiticity for the left and right eigenvectors are discussed in Section 5 after the operator has been constructed explicitly.

3.2 The dynamics as an eigenvalue problem

After the rotation and Trotter decomposition, the two-dimensional network in Figure 1a is contracted along the spatial direction by repeatedly applying the column transfer matrix between the temporal boundaries. For a chain of N spatial sites,

$$\mathcal{L}(T) = \langle L| \mathcal{E}(T)^N |R\rangle = \sum_i c_i \mu_i(T)^N = c_0 \mu_0(T)^N \left[1 + \sum_{i \geq 1} \frac{c_i}{c_0} \left(\frac{\mu_i}{\mu_0} \right)^N \right], \quad (9)$$

where μ_i are the eigenvalues of $\mathcal{E}$ and the coefficients c_i contain the corresponding overlaps with the boundary states. If μ_0 has the largest modulus, every other contribution is suppressed relative to it by $(|\mu_i|/|\mu_0|)^N$. The thermodynamic limit is therefore determined by the leading eigenvalue, and the intensive Loschmidt rate becomes

$$\ell(T) = - \lim_{N \rightarrow \infty} \frac{1}{N} \log |\mathcal{L}(T)| = - \log |\mu_0(T)|, \quad (10)$$

up to $\mathcal{O}(1/N)$ boundary corrections for a finite open chain. The calculation thus reduces to a spectral problem whose cost is independent of the spatial length.

Equation (9) also identifies the spectral data required for the conformal analysis. The modulus of μ_0 determines the Loschmidt rate, its phase encodes the central charge, and the first spectral gap gives access to the boundary operator content. The generalized temporal entropies instead depend on the dominant left and right eigenvectors. Our calculations must therefore resolve the leading eigenpairs of the non-Hermitian operator $\mathcal{E}$.

To obtain these eigenpairs, we implement the block power method described in Section 6. It evolves a block of k temporal states and resolves the k leading eigenvalues together with their corresponding left and right eigenvectors.

Each iteration increases the bond dimension of the temporal states and must therefore be followed by a truncation. The available truncation schemes are introduced in Section 3.4.

3.3 Temporal states and the RTM

Each application of $\mathcal{E}$ increases the bond dimension of the temporal boundary states. The computational cost is therefore controlled by their temporal entanglement, defined across a cut in the time direction. Section 2.2 showed that in CFT the generalized temporal

entropies grow logarithmically with the evolution time. We now define the object from which these entropies are computed.

The dominant left and right fixed points of the non-Hermitian transfer matrix are distinct, and the echo depends on their overlap $\langle L|R \rangle$. An Reduced Density Matrix (RDM) formed from either state alone therefore does not contain the required information. The relevant object is instead the RTM constructed jointly from the two temporal boundaries,

$$\mathcal{T}_t = \frac{\text{Tr}_{>t} |R\rangle \langle L|}{\langle L|R \rangle}, \quad (11)$$

where $\text{Tr}_{>t}$ traces over the $T - t$ temporal sites above the cut and leaves the t sites below it open. Thus, $\mathcal{T}_t$ acts on the first t temporal sites. The denominator normalizes the matrix so that $\text{Tr} \mathcal{T}_t = 1$. Since $\langle L|$ is not the Hermitian conjugate of $|R\rangle$, $\mathcal{T}_t$ is generally non-Hermitian and may have complex eigenvalues. The consequences of this complex spectrum for truncation are discussed in the next subsection. Figure 3 shows the position of $\mathcal{T}_t$ within the strip geometry.

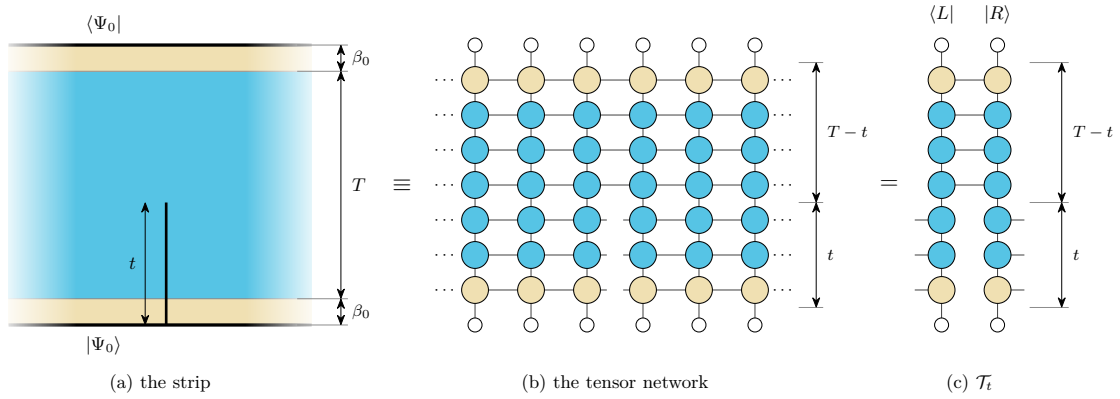


Figure 3: **Position of the reduced transition matrix.** (a) The cut lies on one spatial bond of the strip and extends from the bottom edge to height t . (b) In the tensor-network representation, the horizontal bonds across the cut remain open over the first t temporal sites, while the two sides remain contracted over the remaining $T - t$ sites. (c) Contracting the columns to the left and right of the cut gives $\langle L|$ and $|R\rangle$, respectively. The power method of Section 6 converges these states to the dominant left and right eigenvectors of $\mathcal{E}$. Their contraction over the upper $T - t$ sites gives $\mathcal{T}_t$ in Eq. (11), with the lower t indices left open.

The spectrum of $\mathcal{T}_t$ defines the generalized temporal entropies associated with the transition between the two boundary states. We use the Rényi-2 member of the family introduced in Section 2,

$$S_2^{\text{gen}}(t) = -\log \text{Tr}(\mathcal{T}_t^2). \quad (12)$$

When $\langle L|$ and $|R\rangle$ differ, there is no gauge in which $\mathcal{T}_t$ is locally diagonal, so its spectrum cannot be obtained from a local decomposition. We choose the Rényi-2 entropy because $\text{Tr}(\mathcal{T}_t^2)$ can instead be evaluated directly by contracting two copies of the RTM, without diagonalizing it. This simplification comes with finite-time corrections that must be accounted for when comparing with the conformal predictions, as discussed in Section 2.2.

Once $\langle L|$ and $|R\rangle$ have converged for a target evolution time T , we evaluate Eq. (12) at every internal bond of the temporal chain, corresponding to cuts $t \in (0, T)$. The entropy is generally complex because $\mathcal{T}_t$ is non-Hermitian, so its real and imaginary parts are analysed separately as functions of the normalized cut position.

The temporal region represented by $\mathcal{T}_t$ extends from the lower edge of the strip to the bond at t and therefore has only one endpoint in the interior. Appendix A.1 explains how this edge-anchored geometry fixes the entropy coefficient; if the interval instead had both endpoints in the interior, the logarithmic slope in Eq. (5) would be twice as large. The central charges reported in Section 7 are extracted using the edge-anchored geometry.

3.4 Truncation based on RDM and RTM

We use two complementary truncation schemes based on the **RDM** and the **RTM**. The conventional **RDM** truncation, available in most tensor-network libraries, brings a boundary **MPS** into mixed canonical form and discards the smallest singular values of its orthogonality centre, giving the optimal low-rank approximation to that state [6, 30]. The gauge-invariant scheme implemented in `ITransverse.jl` [28] instead truncates the **RTM** in Eq. (11), which depends on both boundaries. We use the **RTM** scheme for production calculations and retain the **RDM** scheme as a complementary check. Appendix C provides further details on both truncations, including their response to gauge transformations and the conditions for optimal truncation. Here we focus on their main differences: their computational cost and the quantities discarded during truncation.

The **RTM** scheme is substantially cheaper because the two schemes preserve different information. The eigenvalues of the boundary **RDM** are the squared Schmidt coefficients of $|R\rangle$ across the cut. Truncating them retains the directions in which $|R\rangle$ has the greatest weight, irrespective of their overlap with $\langle L|$. The resulting approximation therefore preserves components of $|R\rangle$ that do not contribute to $\langle L|R\rangle$. By contrast, the **RTM** weights each direction by its contribution to this overlap: a large component of $|R\rangle$ carries little weight if it is orthogonal to the corresponding component of $\langle L|$. Thus, the **RTM** can in principle represent the relevant part of both boundaries with fewer retained values and a smaller bond dimension.

The lower cost of the **RTM** scheme comes with a weaker truncation guarantee. For a Hermitian, positive **RDM**, the singular values coincide with the eigenvalues, so retaining the largest Schmidt values gives the optimal approximation at a fixed bond dimension. The **RTM** is instead non-Hermitian, and its eigenvalues may be complex or negative. Although its normalized trace satisfies $\text{Tr } \mathcal{T}_t = \sum_i \lambda_i = 1$, contributions from different eigenvalues may cancel. There is therefore no longer a one-to-one correspondence between the eigenvalues and the singular values. The overlap error can nevertheless be bounded: if the truncation changes the left and right temporal states by at most ϵ_L and ϵ_R in norm, respectively, then the error in their overlap is bounded by $\epsilon_L + \epsilon_R + \epsilon_L \epsilon_R$ [31].

The **RTM** truncation targets contributions to the overlap rather than minimising these norm errors, so the discarded singular values cannot be used directly to evaluate the bound; however, the scheme works in practice for the physical reasons discussed in Appendix C. When its results are noisy or difficult to interpret, we use the more expensive **RDM** scheme as a fallback. Section 7 verifies that the two schemes agree wherever both converge.

4 The ANNNI-type Model

The framework of Sections 2 and 3 requires a quench to a critical point. We study a self-dual quantum spin chain that extends the transverse-field Ising model with **NNN** interactions [20]. For spin-1/2 degrees of freedom, the Hamiltonian is

$$H = - \sum_i \left[\sigma_i^z \sigma_{i+1}^z + \lambda \sigma_i^x + p (\sigma_i^z \sigma_{i+2}^z + \lambda \sigma_i^x \sigma_{i+1}^x) \right], \quad (13)$$

where $\sigma^{x,z}$ are Pauli matrices, λ is the transverse field, and p controls the strength of the additional interactions. The $\sigma^x \sigma^x$ term preserves self-duality under the Kramers–Wannier transformation [20], which fixes the critical point at $\lambda = 1$. We work at this point throughout, so that the conformal predictions of Section 2 apply.

At $p = 0$, Eq. (13) reduces to the integrable transverse-field Ising chain. For $p > 0$, a Jordan–Wigner (JW) transformation no longer maps the Hamiltonian to free fermions, and the model is therefore non-integrable. The origin of the fermionic interaction is detailed in Appendix D.

4.1 Equilibrium criticality and the central charge

Previous finite-size scaling studies find that the chain remains critical and belongs to the Ising universality class up to $p \lesssim 1.5$, with central charge $c \approx 1/2$ [20]. The universality class of this same model has also been established by a different method, which extracts the conformal data directly from the lattice Hamiltonian [32]. We restrict our simulations to $0 \leq p \leq 1.5$, the range over which the model is known to remain critical and in the Ising universality class [20].

We verify the equilibrium universality class over this range using the ground-state entanglement. For a critical open chain of length L , CFT predicts that the von Neumann entropy of a block of length l scales as [24, 30, 33]

$$S_{\text{vN}}(l, L) = \frac{c}{6} W(l, L) + k, \quad (14)$$

where W is the chord variable of Eq. (4) and k is a non-universal constant. The von Neumann entropy is well suited to this calculation because DMRG provides the full ground-state Schmidt spectrum directly, and also has weaker finite-size corrections than the higher-order Rényi entropies [34].

For each value of p , we obtain the ground state of a chain with $N = 400$ sites and evaluate the entropy across every bond. Equation (14) predicts a linear dependence on $W(l, L)$, so the entanglement profile of a single ground state provides an estimate of the central charge. We fit only the middle part of the chain, using the cuts satisfying $0.25L < l < 0.75L$, where boundary corrections are smallest. The resulting profiles are shown in Figure 4.

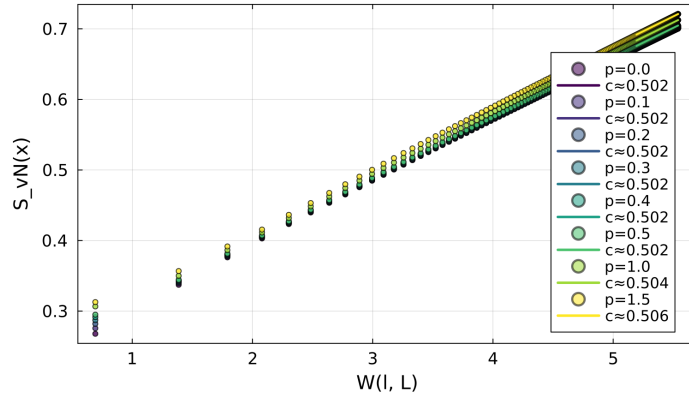


Figure 4: **Effect of NNN interactions to the equilibrium central charge.** Ground-state entanglement profiles at $\lambda = 1$ and $N = 400$, for $p \in [0.0, 1.5]$. The linear dependence on the chord variable W gives a central charge close to the Ising prediction $c = 1/2$ at every p , with a small common offset from finite-size corrections. The fitted central charge ranges from $c = 0.5018$ to 0.5055 .

The profiles follow the conformal scaling law at every coupling, and the fitted central charge barely changes with p . Its small common offset from the exact value $c = 1/2$ is consistent with residual finite-size corrections to Eq. (14), since a similar offset is already present at the exactly solvable point $p = 0$. The additional interactions therefore do not change the universality class over the range considered.

The profiles separate at small W , where the cuts lie closest to the open boundaries. Equation (14) is less accurate in this region because the boundaries introduce corrections that depend on the microscopic interactions and therefore on p . This is why these cuts are excluded from the fit. Towards the centre of the chain, these boundary effects weaken and the profiles converge towards the common linear behaviour predicted by CFT. Appendix E.1 gives an independent estimate from the finite-size scaling of the mid-chain entropy and a cross-check using the Rényi-2 entropy.

4.2 Sound velocity

The NNN interaction modifies the propagation of low-energy excitations. At a critical point, the gap closes and the dispersion $\omega(k)$ becomes linear in the lattice momentum k near the gap-closing point. Its slope defines the sound velocity through $\omega(k) = v|k|$, which sets the relativistic scale of the continuum theory. This velocity enters the conformal predictions of Section 2, so it must be determined independently before they can be compared with the dynamical data. For the integrable chain, the exact dispersion $\omega(k) = 2|\sin k|$ gives $v = 2$ [35]; for $p > 0$, we determine v numerically.

We extract $v(p)$ from the equilibrium dispersion of periodic chains, obtained by exact diagonalization, by identifying the lowest excitation with non-zero momentum. Because exact diagonalization is restricted to finite rings, we extrapolate the velocities obtained from five system sizes using their leading $1/N^2$ finite-size dependence. This procedure reproduces the exact result at $p = 0$ and is then applied at each non-zero coupling p .

The velocity grows steadily from 2.000 at $p = 0$ to 10.799 at $p = 1.5$, as shown in Figure 5. We use the independently extrapolated value at each coupling in all subsequent conversions. Although the velocity is non-universal, it is not a fitted parameter in our analysis: it is fixed by the model and determined from the equilibrium spectrum. Appendix E.2 describes the dispersion calculation, extrapolation and validation in detail.

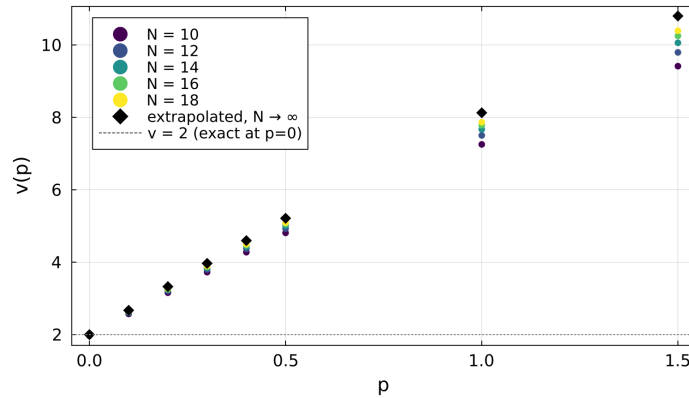


Figure 5: **The sound velocity grows with frustration.** $v(p)$ at $\lambda = 1$, from the equilibrium dispersion of periodic rings of $N = 10$ to 18 sites (coloured circles), together with the values extrapolated to the thermodynamic limit (black diamonds), which are the ones used throughout. The dotted line marks the exact Ising value $v = 2$ at $p = 0$.

5 Building the Time-Evolution MPO

For nearest-neighbour (NN) Hamiltonians, the evolution operator $U(\delta t) = e^{-iH\delta t}$ is usually represented as a Trotter–Suzuki circuit of two-site gates, giving rise to the usual brick-wall structure [13]. However, an interaction between sites i and $i + 2$ cannot be assigned to a single bond within this pattern. To preserve the one-site translation invariance required by the transverse contraction, as discussed in Section 3.1, we instead express the Hamiltonian as a finite-state machine (FSM) and exponentiate the resulting representation directly.

5.1 The Hamiltonian as a FSM

In the FSM representation [36, 37], the Hamiltonian is encoded as an MPO whose local tensor has a block-upper-triangular form [38]. Two identity blocks propagate the initial and final states, while the on-site block D generates local terms. The block row C initiates interactions, the block column B terminates them, and the memory block A propagates unfinished interactions between their endpoints. The virtual index therefore records whether an interaction has not yet begun, is being propagated, or has terminated. For more complicated Hamiltonians, this block structure can be generated automatically from the operator [39]. For the ANNNI-type model, the local tensor has bond dimension $D_W = 5$ and takes the form

$$W_H = \begin{pmatrix} \mathbb{1} & C & D \\ 0 & A & B \\ 0 & 0 & \mathbb{1} \end{pmatrix} = \begin{pmatrix} \mathbb{1} & -\sigma^z & -p\lambda\sigma^x & -p\sigma^z & -\lambda\sigma^x \\ 0 & 0 & 0 & 0 & \sigma^z \\ 0 & 0 & 0 & 0 & \sigma^x \\ 0 & \mathbb{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbb{1} \end{pmatrix}. \quad (15)$$

Figure 6 represents Eq. (15) through the allowed paths between the five virtual states. Each node denotes a value of the virtual index, while an arrow from state a to state b , labelled by an operator O , represents the tensor entry $(W_H)_{ab} = O$. The FSM moves from left to right along the chain, beginning in state 1 and ending in state 5. The loop $1 \rightarrow 1$ applies identities until a Hamiltonian term is initiated. A direct transition $1 \rightarrow 5$ applies the on-site field and completes the term at the same site. An NN interaction instead follows either $1 \rightarrow 2 \rightarrow 5$ or $1 \rightarrow 3 \rightarrow 5$: the first transition applies the operator at site i and records whether a σ^z or σ^x endpoint remains to be applied; the second applies the corresponding operator at site $i + 1$ and completes the interaction. The NNN term follows $1 \rightarrow 4 \rightarrow 2 \rightarrow 5$: the first transition applies σ^z at site i , the second inserts an identity at site $i + 1$, and the third applies σ^z at site $i + 2$. After state 5 is reached, the loop $5 \rightarrow 5$ applies identities along the remainder of the chain. Zero entries in W_H correspond to absent arrows, so no other transitions are allowed.

Because the FSM is directional, its local tensor is asymmetric under interchange of the virtual indices. The exponentiated operator inherits this structure, as does the spatial transfer matrix after moving to the transverse picture. The resulting operator $\mathcal{E}$ is generally non-Hermitian and non-normal, $\mathcal{E}\mathcal{E}^\dagger \neq \mathcal{E}^\dagger\mathcal{E}$. Its left and right eigenvectors are therefore not related by Hermitian conjugation, and we compute the two temporal boundary states separately.

5.2 Exponentiating the FSM

We seek an approximation $W(\tau) \approx e^{\tau H}$, with $\tau = -i\delta t$, constructed from the blocks of W_H . Neither of the most direct approaches gives a suitable many-body operator: exponentiating

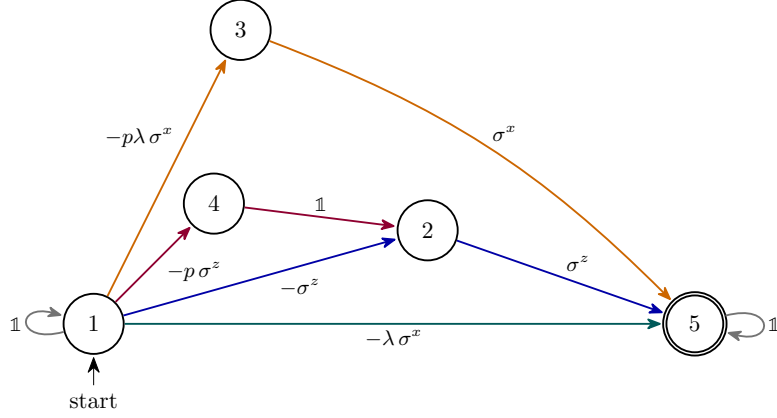


Figure 6: **The Hamiltonian as a finite-state machine (FSM)**. The nodes are the five virtual states of Eq. (15); each arrow is labelled by the operator applied during the transition, and every path from state 1 to state 5 (double circle) generates one term of the Hamiltonian. The arrows are coloured according to the generated term: the transverse field in teal, the **NN** couplings in blue ($\sigma^z\sigma^z$) and orange ($\sigma^x\sigma^x$), and the **NNN** coupling in purple. The grey loops apply identities before and after the Hamiltonian term. State 2 is shared by the blue and purple paths because both terminate with σ^z , keeping the bond dimension at $D_W = 5$.

the local tensor as an ordinary matrix mixes its operator-valued blocks but does not reproduce the exponential of the full Hamiltonian; on the other hand, a finite-order Taylor expansion of $e^{\tau H}$ avoids this local exponentiation, but it is not size-extensive because H^n contains $\mathcal{O}(N^n)$ contributions [40].

We instead use a construction that organises the expansion according to the spatial overlap of the local Hamiltonian terms [40]. Contributions with disjoint support commute and factorise, allowing them to be retained to all orders. Only connected clusters of overlapping terms must be reproduced explicitly. We use the second-order member of this family, denoted *VD2*, whose virtual bond dimension is 13 for the model of Eq. (13). Appendix F gives the full construction, the explicit tensor and the origin of this bond dimension.

Cheaper implementations—with lower bond dimension—include the operators W^I and W^{II} [41]. For a strictly **NN** Hamiltonian, W^{II} is effectively second order. In the present model, however, it omits connected clusters generated by the **NNN** interaction and consequently provides only first-order accuracy. Appendix F.4 demonstrates this difference. We therefore use *VD2* despite its larger bond dimension.

After moving to the transverse picture, the virtual bond dimension of the evolution **MPO** becomes the local physical dimension of the temporal chain. The *VD2* construction therefore has a larger local dimension than W^{II} . Its second-order accuracy permits the larger time step $\delta t = 0.1$ for a given discretisation error, which reduces both the temporal length at fixed physical time and the number of required truncation steps, making the long-time calculations of Section 7 more computationally accessible with *VD2*.

6 The Block Power Method

Section 3 reduced the dynamical calculation to finding the leading eigenpairs of the transfer matrix $\mathcal{E}$. We obtain them with a block power method that evolves several temporal states simultaneously. The method returns the dominant eigenvalues and their left and right eigenvectors, which we then identify and order for comparison with the **CFT** predictions.

Appendix G gives the implementation details.

The right and left eigenvectors of $\mathcal{E}$ satisfy

$$\mathcal{E} |R_i\rangle = \mu_i |R_i\rangle, \quad \langle L_i| \mathcal{E} = \mu_i \langle L_i|. \quad (16)$$

Because $\mathcal{E}$ is non-normal (Section 5.1), its left and right eigenvectors differ, and neither set is orthogonal by itself. Instead, the two sets are biorthogonal, $\langle L_i | R_j \rangle \propto \delta_{ij}$. Here the bra notation does not imply complex conjugation: $\langle L_i |$ is a row vector, and $\langle L_i | R_j \rangle$ denotes the direct contraction of the two temporal networks. We therefore propagate the left states with the transpose $\mathcal{E}^\top$, which implements the second equation in Eq. (16).

Different outputs of the calculation require different parts of this eigensystem. The modulus of the dominant eigenvalue μ_0 determines the Loschmidt rate, while its phase encodes the central charge. Phase gaps to the subleading eigenvalues contain the boundary operator content, and the dominant pair $(\langle L_0 |, |R_0\rangle)$ defines the generalized temporal entropies. We must therefore resolve the leading eigenvalues individually, identify μ_0 , and order the remaining eigenvalues for comparison with the conformal tower.

6.1 The algorithm

A single-vector power method is sufficient when only the dominant eigenpair is required. It therefore provides the Loschmidt rate, the central charge obtained from the phase of μ_0 , and the dominant left and right states used in the entropy calculation. Extracting the boundary operator content also requires the subleading eigenvalues, whose separations decrease as T grows. We resolve these eigenvalues by iterating a block of k states. The block converges to the leading k -dimensional invariant subspace at a rate controlled by $|\mu_{k+1}/\mu_k|$; near-degeneracies among eigenvalues within the block therefore do not slow the convergence of the subspace. The projected eigenproblem then resolves the individual eigenpairs within this subspace.

The block method evolves k right states $|R_j\rangle$ and k left states $\langle L_j|$, each represented as a temporal Matrix Product State (tMPS). We initialise the two blocks independently with random complex tensors. Algorithm 1 summarises one iteration, and Figure 7 shows the tensor-network operations used to apply and project the transfer matrix.

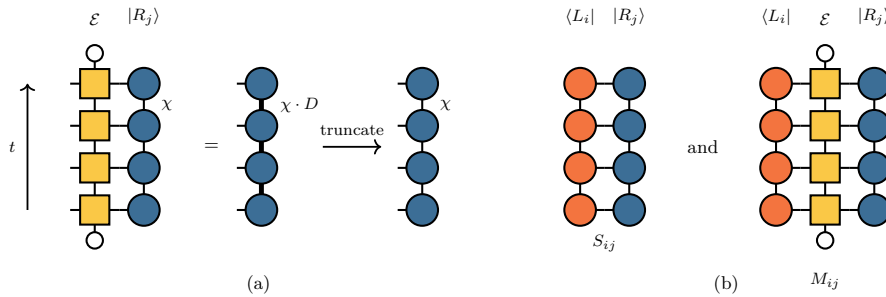


Figure 7: **One iteration of the block power method.** Each tMPS forms a column, while $\mathcal{E}$ is the corresponding column of the space–time network closed at both ends by the product state (small open circles). (a) Applying $\mathcal{E}$ to one member of the right block, as in Eq. (17), increases the temporal bond dimension from χ to χD , where D is the tMPO bond dimension. The result is then truncated back to χ . The left block is treated analogously using the transpose. (b) Contractions defining the two $k \times k$ matrices in Eq. (19). The left chain enters directly as a bra, without complex conjugation.

Algorithm 1 Block power method for the k dominant eigenpairs of $\mathcal{E}$.

- 1: $|R_j\rangle, \langle L_j| \leftarrow$ random complex **tMPS**s, $j = 1, \dots, k$
 - 2: **repeat**
 - 3: $|\tilde{R}_j\rangle \leftarrow \text{truncate}_\chi(\mathcal{E}|R_j\rangle), \quad \langle \tilde{L}_j| \leftarrow \text{truncate}_\chi(\langle L_j| \mathcal{E}) \quad \triangleright$ apply
 - 4: $S_{ij} \leftarrow \langle L_i|R_j\rangle, \quad M_{ij} \leftarrow \langle L_i|\tilde{R}_j\rangle = \langle L_i|\mathcal{E}|R_j\rangle \quad \triangleright$ project
 - 5: solve $Mv = \theta Sv$ for Ritz values $\theta_1, \dots, \theta_k$ and coefficients V, U
 - 6: $|R_j\rangle \leftarrow \text{truncate}_\chi(\sum_i V_{ij}|\tilde{R}_i\rangle), \quad \langle L_j| \leftarrow \text{truncate}_\chi(\sum_i U_{ij}\langle \tilde{L}_i|) \quad \triangleright$ de-mix
 - 7: normalise every block member
 - 8: **until** the leading Ritz values stop changing within tolerance
-

The first step applies the transfer matrix to each member of the right block and its transpose to each member of the left block,

$$|\tilde{R}_j\rangle = \mathcal{E}|R_j\rangle, \quad \langle \tilde{L}_j| = \langle L_j| \mathcal{E} \iff |\tilde{L}_j\rangle = \mathcal{E}^\top |L_j\rangle. \quad (17)$$

Applying the **tMPO** multiplies the bond dimension of each state by D , as shown in Figure 7a. After each application, we use either scheme from Section 3.4 to truncate the propagated state to the maximum bond dimension χ , and then normalise it. This truncation bounds the cost of subsequent iterations.

The next step projects the eigenproblem onto the current block subspaces. We approximate a right eigenvector by

$$|\psi\rangle = \sum_{j=1}^k v_j |R_j\rangle. \quad (18)$$

Inserting this ansatz into $\mathcal{E}|\psi\rangle = \theta|\psi\rangle$ produces a residual. We require its contraction with every member of the left block to vanish, so that $\langle L_i|(\mathcal{E} - \theta)|\psi\rangle = 0$. This condition reduces the full eigenproblem to the two $k \times k$ matrices shown in Figure 7b,

$$S_{ij} = \langle L_i|R_j\rangle, \quad M_{ij} = \langle L_i|\mathcal{E}|R_j\rangle, \quad (19)$$

and the generalized eigenvalue problem

$$\sum_j M_{ij} v_j = \theta \sum_j S_{ij} v_j. \quad (20)$$

The overlap matrix S describes the relative orientation of the left and right subspaces, while M is the transfer matrix projected between them. Solving Eq. (20) yields the Ritz values θ_j , which approximate $\mu_0, \dots, \mu_{k-1}$. It also yields the coefficient matrices V and U used in Algorithm 1. The columns of V specify the combinations of propagated right states associated with the approximate right eigenvectors, and U provides the corresponding combinations for the left block. If S is ill-conditioned, the oblique projection amplifies numerical errors. We therefore use the pseudoinverse regularisation described in Appendix G when solving Eq. (20).

We then use the Ritz coefficient matrices to separate the propagated blocks into approximate eigenpairs. Before the next iteration, we truncate the resulting states back to χ and normalise them. The iteration stops once the leading Ritz values remain stable within the prescribed tolerance.

The tensor-network operations dominate the computational cost, whereas solving the $k \times k$ eigenproblem is negligible. Most of the cost arises from the sums used to separate

the propagated block states. Appendix G.4 quantifies these contributions and describes the measures used to reduce them.

When only the eigenvalues are required, we retain the blocks as orthonormal bases of the leading invariant subspaces instead of assigning individual members to separate eigenvectors. This representation is better conditioned near degeneracies and does not change the Ritz values, as discussed in Appendix G.2. We use it for the late-time spectral measurements in Section 7.

Whenever $\mathcal{E}$ is small enough to construct explicitly, we compare the block method with exact diagonalisation. The leading eigenvalues agree with this benchmark, validating the numerical pipeline (Appendix G.3).

6.2 Selecting and tracking the physical branch

At each evolution time, the block method returns k Ritz values from which we must identify the physical eigenvalue μ_0 . At the first time point, we select the value of largest modulus. This criterion is unambiguous at small T , where the leading spectral gap is wide. At subsequent times, we again select the value of largest modulus unless the two leading candidates agree within one per cent. In that case, we choose the candidate closest in the complex plane to the previously accepted μ_0 . Since the tied candidates have almost equal moduli, this criterion reduces in practice to continuity of the phase. Such near-ties are expected from the conformal spectrum: as T increases, the moduli of the states in the conformal tower approach a common value because of the emergent dual unitarity discussed in Section 2.2. We then order the subleading eigenvalues by their phase distance from μ_0 , following the ordering of the conformal tower.

For a sequence of evolution times, the converged block at one time provides the initial state for the next. Consecutive calculations therefore begin from the same leading subspace, which helps the selection remain on a continuous spectral branch as the eigenvalues become closer. If the block contains too few tower states to resolve the required operator content, we increase its size in steps, $k = 4 \rightarrow 6 \rightarrow 8$. Each larger calculation is initialised from the converged smaller block, so the additional cost is incurred only when the spectrum requires it. Appendices G.6 and G.7 describe the selection safeguards and conditioning diagnostics used throughout this procedure.

7 Results

The four predictions of Section 2.2 are now confronted with the simulations. All of them are measured on the ANNNI-type chain of Eq. (13) at $\lambda = 1$, quenched from $|X+\rangle^{\otimes N}$, with the frustration p tuned from the integrable Ising point $p = 0$ up to $p = 1.5$. A window, in what follows, is the interval of evolution times over which a quantity can be fitted. It opens once T is long enough for the conformal asymptotics to hold, and closes when the numerics stop resolving the object being fitted. The predictions split into two families, and this section follows that split. The entropy route goes through the eigenvectors: the generalized Rényi-2 entropy of Eq. (12) is compared with the conformal profile of Eq. (5). The spectral route goes through the eigenvalues alone. It carries three of the predictions: emergent dual unitarity, the central charge from the phase curvature of λ_0 (Eq. (6)), and the boundary exponent from the first gap (Eq. (7)). In each route the integrable point is measured first, where every prediction has an exact target [17], and frustration is switched on afterwards.

The two routes behave very differently once the model stops being integrable, and

that difference organises this section. The entropy route needs both the left and the right eigenvector of the transfer matrix. Its window closes early at $p \neq 0$, and no amount of extra evolution time reopens it. The spectral route needs only the eigenvalues, which are Rayleigh quotients and therefore accurate to second order in the eigenvector error. It survives frustration and carries the measurement of the central charge across the whole range studied.

Unless stated otherwise, all runs use the *VD2* time-evolution MPO of Section 5 with $\delta t = 0.1$, the regulator $\beta_0 = 0.2$ ($N_\beta = 4$, Section 2.1), and the initial state $|X+\rangle$. At $p = 1.0$ and $p = 1.5$ the Trotter step is halved and quartered, with N_β raised to keep $\beta_0 = 0.2$, because the norm drift after eight *VD2* layers reaches 1.56 and 6.96 at $\delta t = 0.1$ (Appendix F.4). The temporal states come from the power method of Section 6, at maximum bond dimension $\chi = 64$. Every eigenvalue-based measurement is converted into universal data with the extrapolated sound velocity $v_\infty(p)$ of Section 4.2, and the non-universal prefactors are removed through ratios or the calibration against the integrable point, following Appendix B. The fit conventions, and the controls that bound how far each number can move, are set out in Appendix E.3.

7.1 The entropy route

At $p = 0$ the generalized temporal entropy follows the conformal prediction over the full simulated window. Figure 8 shows the Rényi-2 profiles: the real part collapses onto the chord curves of Eq. (5) with $c = \frac{1}{2}$, and the imaginary part forms the predicted plateau.

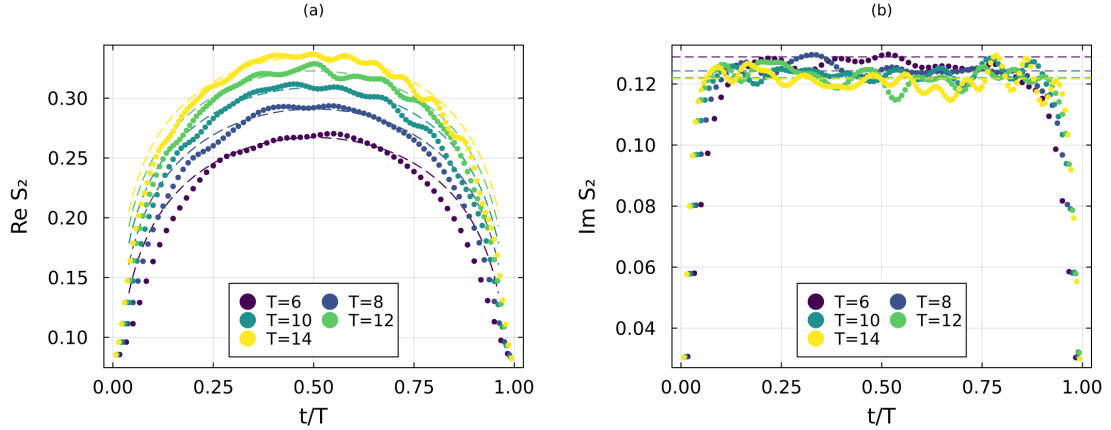


Figure 8: **The generalized temporal entropy at the integrable point.** One colour per evolution time. (a) $\text{Re } S_2$ against t/T . Each dashed curve is the prediction of Eq. (5) at $c = \frac{1}{2}$ for that T , not a fit: its shape is fixed by the prediction, and only its height is matched to the data, since the additive constant s_0 is non-universal. What is being compared is therefore the shape. (b) $\text{Im } S_2$, flat across the interior of the strip as predicted. Each dashed line here is the measured plateau of its own T . The plateaus fall from 0.129 at $T = 6$ to 0.122 at $T = 14$, approaching the predicted $\pi c/16 = 0.098$ from above as the finite-time correction decays.

The plateau sits above $\pi c/16$ at every time we reach and falls as T grows. Whether that fall converges on the predicted value is a question about the finite-time correction, and answering it needs a long ladder. Over $T = 2$ to 11 the plateau looks flat at 0.12, and a constant, a $T^{-1/2}$ decay and a $1/T$ decay are indistinguishable: all three extrapolate to $c \approx 0.69$, which is wrong by nearly 40%. Extending the same ladder to $T = 24$ separates them. The $T^{-1/2}$ form predicted in Ref. [19] then extrapolates the plateau to 0.0994 against the exact $\pi/32 = 0.0982$, and the constant model is rejected against it by a

difference in corrected Akaike information of 21. Repeating the whole ladder at $\delta t = 0.05$ moves the extrapolation by less than the spread across fit windows. The absolute entropy route therefore returns

$$c = 0.51 \quad (p = 0), \quad (21)$$

against the exact $\frac{1}{2}$. This is an absolute measurement: it uses no reference model and no calibration. The same comparison in Ref. [19], on a different implementation, returns 0.518 with the correction and 0.67 without it, so both the corrected value and the size of the uncorrected bias agree with what we find here.

The chord slope of the real part is the less useful of the two estimators. Doubling the bond dimension leaves the plateau unchanged to about 2% but moves the chord slope by up to 0.12 at the same evolution time, so the slope carries a truncation systematic as large as the effect being measured. Every entropy number quoted here is therefore read from the imaginary plateau.

With frustration the window closes quickly. Table 1 lists the last evolution time at which the plateau still falls monotonically towards its target; beyond it the plateau turns back upwards, which is the sharpest indicator we have that a rung has stopped being meaningful. The window shrinks from $T \approx 20$ at $p = 0$ to $T \approx 4$ at $p = 0.5$.

At $p = 0.1$ the window is still long enough to separate the correction laws. Fitting the plateau over $T = 2$ to 14 prefers the $T^{-1/2}$ form over a constant by a difference in corrected Akaike information of 9.6, and extrapolates to $c = 0.52$. Two of those rungs, at $T = 12$ and $T = 14$, lie beyond the single-run window and were obtained by repeating the calculation with independent seeds until two runs agreed, as described in Section 7.1. The rung at $T = 13$ is excluded: its plateau of 0.184 breaks the smooth variation of its neighbours at 0.116 and 0.104, and retaining it reverses the model preference. That sensitivity is the honest measure of how much this extrapolation rests on a single rung.

From $p = 0.3$ upwards the window is too short for the comparison to decide anything: the four models extrapolate to anywhere between 0.37 and 1.25. Beyond $p = 0.1$ the absolute entropy route therefore gives no central charge, and the spectral route of Section 7.2 carries the measurement.

p	last usable T	phase rigidity there	rigidity at $T = 14$
0	≈ 20	2.6×10^{-6}	1.3×10^{-4}
0.1	11	2.0×10^{-8}	1.6×10^{-10}
0.3	7	3.0×10^{-6}	—
0.5	4	2.2×10^{-4}	—

Table 1: **Where the entropy route stops.** The window is set by the turning point of the imaginary plateau. The phase rigidity at that point spans four orders of magnitude across couplings, so there is no single value of the rigidity at which the route fails.

Why the window closes is worth stating carefully, because the obvious explanation does not survive inspection. The phase rigidity of the dominant pair falls geometrically with T at every coupling, and much faster with frustration: at $T = 14$ it is 1.3×10^{-4} at $p = 0$ and 1.6×10^{-10} at $p = 0.1$. It is tempting to read the closing window as the direct consequence. Two observations argue against that reading. The rigidity at the last usable time spans four orders of magnitude across couplings (Table 1), so there is no threshold value at which the method fails. And the rigidity falls monotonically while the failures are intermittent: at $p = 0$ the rungs at $T = 17$ and $T = 22$ are unusable while those at $T = 18$ to 21 and at $T = 23$ are not. A monotonic quantity cannot by itself produce intermittent failure.

The power method starts from a random complex seed, which suggests the missing ingredient. Repeating a failed rung with independent seeds recovers it: at $p = 0$, $T = 17$ the ladder returned a plateau of 0.72, and independent repetitions of the same calculation return 0.117 and 0.113, in family with their neighbours. The failure is therefore a property of the particular fixed point the iteration reached, not of the evolution time. Low rigidity is best understood as raising the probability of such a failure rather than as causing it directly: the left and right vectors become nearly orthogonal, many directions become nearly degenerate, and the iteration has more ways to settle on the wrong one. That reading is consistent with everything measured here, but the mechanism has not been demonstrated, and we do not claim it. What is established is narrower: the failures are seed-dependent and can be detected and repeated away, and the window in Table 1 is the range over which a single run can be trusted without repetition.

Repetition is therefore used as a measurement procedure past that range. A rung is computed from independent random seeds until two runs agree on the plateau and on $|\mu_0|$, and the agreed value is kept. Agreement between runs is the only acceptance criterion; the result is never compared with the conformal prediction, since accepting rungs for looking conformal would build the answer into the method. Reproducibility alone is not sufficient, and the $T = 13$ rung at $p = 0.1$ shows why: two seeds agreed there on a plateau of 0.184, far from the values on either side. A second requirement is therefore imposed, that the plateau vary smoothly from one rung to the next, which uses only the smoothness of the quantity in T and not its predicted value.

7.2 The spectral route

The eigenvalue-only mode of Section 6.1 carries the spectral measurements to $T = 20$ at $p = 0$. Figure 9a traces λ_0 in the complex plane over that whole ladder: the modulus is constant at 1.487 while the phase winds, which is the emergent dual unitarity of Section 2.2. The same constancy holds at every coupling studied, with the modulus stable to a few parts in 10^4 across the fitted window.

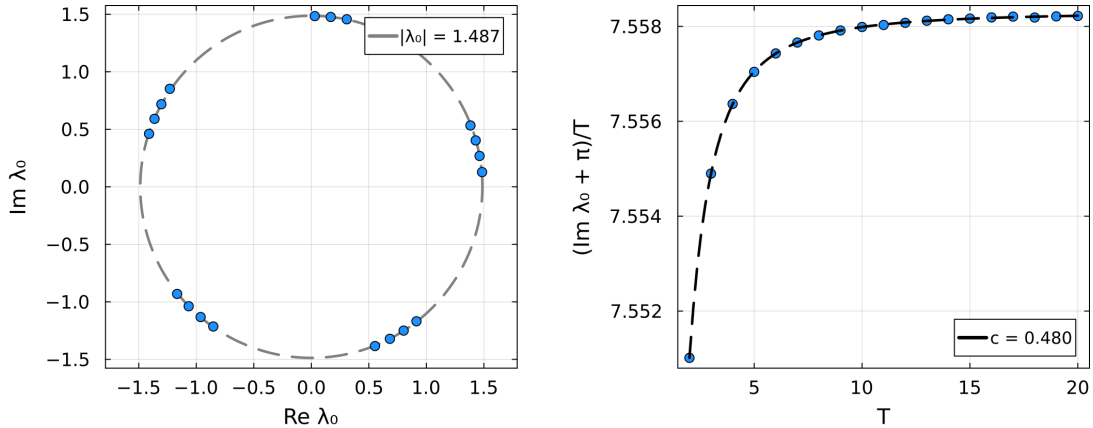


Figure 9: **The spectral predictions at the integrable point, $T = 2$ –20.** (a) λ_0 in the complex plane: constant modulus, winding phase. (b) The fit of Eq. (6) to $\text{Im } \lambda_0$, plotted after subtracting the fitted extensive term $a_0 T$ so that the small Casimir contribution is visible.

The central charge follows from the curvature of the same phase through Eq. (6). Two practical points govern the fit. The phase must be unwrapped before fitting, since $\text{Im } \lambda_0$ advances by between 1.3 and 1.9 radians per unit of evolution time depending on the

coupling and is otherwise returned folded into $(-\pi, \pi]$. And the ladder must be sampled finely enough: what limits these fits is the number of points inside the window, not the reach in T . At $p = 0.3$ a ladder of integer times gives $c = 0.41$ with residuals of 1.2×10^{-3} , while adding half-integer times over the same range gives $c = 0.51$ with residuals of 3.7×10^{-4} .

The fit returns the branch constant as a by-product, and it is a useful check. Left free, it converges to $B/\pi = -1.000$ at every coupling, matching the shift expected from the definition $\lambda_0 = \log(-\mu_0)$. The constant is known, so it is pinned in the quoted fits; where the window is long enough to determine it independently, pinning and leaving it free agree.

Table 2 collects the result. The central charge stays at the Ising value across the whole range of frustration studied, including the two couplings beyond $p = 0.5$. At those two the equilibrium chord fit of Section 4.1 gives 0.5036 and 0.5055, so the dynamical and equilibrium measurements agree to about 1% by two independent routes.

The resolution the measurement demands tightens as p grows, which is why the windows in Table 2 shrink. Every boundary dimension is read from a phase gap of order $\pi x/(vT)$, so raising p from 0 to 1.5 raises the velocity from 2.00 to 10.80 and compresses the same operator content into a phase window five times narrower. The same factor enters the central charge, which is proportional to v : an error in the fitted curvature is amplified five times harder at $p = 1.5$ than at $p = 0$.

p	v_∞	window	spectral c	x_1	entropy-route c
0	2.000	$T \leq 20$	0.45	0.496	0.51
0.1	2.670	$T \leq 16$	0.47	0.497	0.52
0.3	3.967	$T \leq 9$	0.51	0.495	—
0.5	5.212	$T \leq 4.5$	0.53	0.497	—
1.0	8.127	$T \leq 2.3$	0.50	0.54	—
1.5	10.799	$T \leq 2.2$	0.49	0.55	—

Table 2: **Universal data versus frustration.** The spectral central charge comes from Eq. (6) on ladders sampled at $\Delta T \leq 0.5$ inside the window, with the branch constant pinned. The boundary exponent is the first phase gap converted through Eq. (7). Appendix E.3 gives the fit windows and the controls behind these numbers. The entropy route gives an absolute central charge at $p = 0$ and $p = 0.1$; beyond that its window is too short, for the reason given in Section 7.1.

The boundary exponent distinguishes the two boundary conditions of Section 2.2. At the integrable point, fitting Eq. (7) to the first spectral gap over $T = 2-9$ gives $x_1 = 0.496$ for the free boundary ($|X+\rangle$) and $x_1 = 1.996$ for the fixed one ($|Up\rangle$), against the exact values $\frac{1}{2}$ and 2 (Figure 10). With frustration the same measurement returns the ordinary Ising boundary tower. At $p = 0.3$ the first three dimensions are 0.495, 1.577 and 2.181 at $T = 2$, moving to 0.494, 1.555 and 2.105 at $T = 3$; at $p = 0.5$ they are 0.497, 1.591 and 2.214. The leading dimension is at $\frac{1}{2}$ in both cases, and the higher members approach $\frac{3}{2}$ and 2 from above as the evolution time grows.

The same measurement at $p = 1.0$ and $p = 1.5$ gives $x_1 = 0.537$ and 0.548, with the second tower member at 1.392 and 1.481. The tower is therefore still the Ising one at the strongest frustration studied. Both of those rungs sit at $T = 1.2$, which is the shortest time anywhere in this work, and the 8 to 10% excess in x_1 is what the approach to the conformal limit looks like there: at fixed T the deviation falls with p , from 2.3% at $p = 0.1$ to 0.5% at $p = 0.3$ and 0.3% at $p = 0.5$ at $T = 2$, which follows the growth of the velocity, but below $T \approx 2$ no coupling is yet asymptotic. The third member is not resolved at either coupling with $k = 6$.

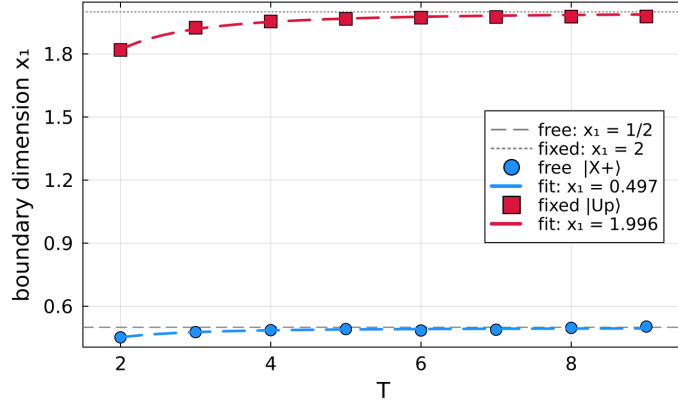


Figure 10: **The boundary exponent at the integrable point.** The first spectral gap converted into a dimension, $x_1 = v T \text{Im}(\lambda_1 - \lambda_0)/\pi$, against T for the free and fixed boundary states. Grey lines mark the exact values $\frac{1}{2}$ and 2. Both approach their target from below as T grows; the fits of Eq. (7) over the whole window give 0.496 and 1.996.

8 Conclusions and Outlook

Conclusions yet to be written.

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A Conformal-field-theory derivations

This appendix collects the standard conformal-field-theory results used in Section 2—the chord form of the two-point function, the strip quantization behind the transfer-matrix spectrum, and the entropy coefficients—and sketches where each comes from. Full derivations are in Refs. [23, 24]; what is specific to this thesis is only how the results are used.

A.1 Conformal maps, the two-point function and the entropy coefficients

In two dimensions the Euclidean coordinates are packaged into $z = x + i\tau$ and $\bar{z} = x - i\tau$. A primary operator ϕ carries a pair of weights $(h, \bar{h})$, its exponents under the two chiral halves of the conformal group, and the combination

$$x = h + \bar{h} \quad (22)$$

is its scaling dimension, the exponent quoted in the main text. On the plane the two-point function is the power law

$$\langle \phi(z_1) \phi(z_2) \rangle = \frac{1}{(z_1 - z_2)^{2h} (\bar{z}_1 - \bar{z}_2)^{2\bar{h}}}. \quad (23)$$

Because every analytic map is a conformal symmetry, this fixes the correlator on any geometry such a map reaches. Mapping the plane onto a cylinder of circumference β with the logarithm map turns the power law into [23, 24]

$$\langle \phi(\omega_1) \phi(\omega_2) \rangle \propto \left[\sinh \frac{\pi(\omega_1 - \omega_2)}{\beta} \right]^{-2h} \left[\sinh \frac{\pi(\bar{\omega}_1 - \bar{\omega}_2)}{\beta} \right]^{-2\bar{h}}. \quad (24)$$

Continuing to real time, with $\beta \rightarrow iT$ and $\sinh \rightarrow \sin$, produces the strip geometry of the quench, and the time-like version of Eq. (24) is the origin of the chord variable of Eq. (4).

The same two-point function fixes the entropy coefficients, which is why the two derivations belong together. The entanglement entropy of an interval $\mathcal{A}$ follows from the Rényi moments $\text{Tr } \rho_{\mathcal{A}}^n$, continued to $n \rightarrow 1$. The replica trick [24] computes those moments by inserting one twist operator at each endpoint of $\mathcal{A}$ that lies in the interior of the system; the twist operator is a primary of dimension [24]

$$\Delta_n = \frac{c}{12} \left(n - \frac{1}{n} \right). \quad (25)$$

For a bulk interval both endpoints are interior, the moments follow the two-point power law $\text{Tr } \rho_{\mathcal{A}}^n \sim \ell^{-4\Delta_n}$, and the entropy is

$$S_n = \frac{c}{6} \left(1 + \frac{1}{n} \right) \log \ell + \text{const}, \quad S_{n \rightarrow 1} = \frac{c}{3} \log \ell + \text{const}. \quad (26)$$

The cuts of this thesis are the other case. Each temporal region is anchored at an edge of the strip, so it shares only one border with its complement: a single interior endpoint, a single twist operator, and half the logarithmic coefficient. That gives $\frac{c}{12} (1 + \frac{1}{n}) - c/6$ at $n \rightarrow 1$ and $c/8$ at $n = 2$ —rather than the bulk $c/3$, and using the bulk coefficient by mistake would double every fitted central charge.

A.2 The transfer-matrix spectrum from the stress tensor

The central charge is carried by the stress-energy tensor, the operator that measures the response of the theory to a change of coordinates. In two dimensions those changes are the symmetry itself, so the modes L_n of the stress tensor generate the conformal transformations; their commutators close into the Virasoro algebra, whose central term is proportional to c , which is where the central charge gets its name [23]. One mode is special: L_0 generates scalings, and its eigenvalues on the primary operators are their weights, so its spectrum is the operator content of the theory.

Quantizing on a strip of transverse size β makes the anomaly of the main text explicit. The generator of translations along the strip is an integral of the stress tensor, and the conformal map from the plane leaves behind one extra, purely geometric term proportional to c [17],

$$\hat{H} = \frac{\pi}{\beta} \left(L_0 - \frac{c}{24} \right). \quad (27)$$

This single expression contains both kinds of universal data: the constant $-c/24$ is the Casimir anomaly and carries the central charge, while the spectrum of L_0 lists the scaling dimensions. The transfer matrix advances one column of the network, so it is $\mathcal{E} = e^{-a\hat{H}}$ with a the spatial lattice spacing, and substituting Eq. (27) fixes its whole spectrum in the form of Eq. (2), with $g = a/\beta$ and $a = 1$ in lattice units: a dominant eigenvalue carrying the Casimir term, and a tower of subleading ones whose ratios and phases carry the operator dimensions.

The width entering on the lattice is not purely Euclidean, since the regulator supplies imaginary time at the two ends and the quench real time in between, and the velocity converts both durations into lengths,

$$\beta = v(2\beta_0 + iT), \quad \frac{1}{\beta} \simeq \frac{1}{v} \left(\frac{2\beta_0}{T^2} - \frac{i}{T} \right), \quad (28)$$

the expansion holding to first order in β_0/T . Substituting it into Eq. (2) gives the eigenvalues of the transfer matrix,

$$\log \mu_i \simeq \frac{i\pi}{vT} \left(x_i - \frac{c}{24} \right) - \frac{2\pi\beta_0}{vT^2} \left(x_i - \frac{c}{24} \right). \quad (29)$$

The universal data therefore appear in the phases at order $1/T$, and in the moduli only through the regulator, at order β_0/T^2 . Taking $x_i = 0$ for the identity gives Eq. (6), and the difference between two eigenvalues gives Eq. (7). The moduli are flat to the extent that β_0 is small, which is the emergent dual unitarity measured in Section 7.

B Universal versus non-universal constants

The predictions of Section 2 are exact only in the continuum limit. A simulation instead runs on a microscopic lattice, at finite evolution time T , bond dimension χ and Trotter step δt , and with a nonzero regulator β_0 . Its measurements therefore mix two kinds of contributions. The universal ones are fixed by the universality class: the central charge c and the scaling dimensions x_i , which set every coefficient in Eqs. (5)–(8). The non-universal ones are introduced by the lattice and the algorithm: the regulator β_0 and the entropy offset s_0 it feeds, the Trotter step δt , the sound velocity v , and the overall normalization of the transfer matrix, which is model dependent and makes the modulus of the dominant eigenvalue differ from one. How well these contributions can be separated

determines which predictions can be compared with a simulation and how that comparison should be made.

The sound velocity is one of these non-universal constants, and we have already determined it: the extrapolated $v_\infty(p)$ of Section 4.2 is the velocity entering every conformal formula below. Since the eigenvalue phases depend on the combination $\pi x/(vT)$, it is what converts a measured phase gap into the corresponding universal scaling dimension x_i .

This separation is possible because the universal and non-universal contributions affect the spectrum differently. The transfer matrix carries an overall normalization that the lattice construction fixes and the continuum theory does not. Each column of the network is one Trotter layer together with the cooling rows, so its scale depends on δt , on β_0 , and on the normalization convention chosen for the local MPO tensor. Its closed form is not needed: the constant multiplies every eigenvalue alike, so it rescales all the moduli $|\mu_i|$ and leaves the phases untouched. The universal level structure therefore appears in the phases, and we read scaling dimensions from phase differences, using the moduli only through ratios or their limiting behaviour.

Ratios remove more than the normalization. A single phase gap contains a universal dimension, but also depends on the regulator β_0 and the velocity. The ratio of two gaps,

$$\frac{\text{Im}(\lambda_2 - \lambda_0)}{\text{Im}(\lambda_1 - \lambda_0)} = \frac{x_2 - x_0}{x_1 - x_0}, \quad (30)$$

is independent of all three at once, since β_0 , the velocity and the normalization cancel between numerator and denominator. It therefore probes the operator content without any external input and provides the cleanest measurement available to us.

Not every quantity can be turned into a ratio of this kind. The entropy slope of Eq. (5) measures c directly, but at finite $(T, \chi, \delta t)$ the measured slope is not the continuum coefficient,

$$\text{slope}(p) = \frac{c(p)}{8} K(T, \chi, \delta t, \beta_0) + R(p; T), \quad (31)$$

where K collects the finite-time, truncation and Trotter corrections that two models in the same universality class share when run through the same pipeline at matched parameters, and R collects the subleading corrections they do not share: their amplitudes involve the operator content and boundary amplitudes of the specific model [19], so they depend on the coupling. The integrable point of the chain makes the shared factor measurable: there $c = \frac{1}{2}$ exactly, so the measured $\text{slope}(0)$ determines K itself. Dividing the two slopes then cancels K , though not R ,

$$c(p) \approx \frac{1}{2} \frac{\text{slope}(p)}{\text{slope}(0)}. \quad (32)$$

The cancellation therefore assumes that the two models share a universality class, which is part of what the measurement is meant to establish, and the residual R sets the uncertainty of the calibrated number, which is why it is a consistency check rather than an independent determination. The second route uses the fact that the functional form of the corrections is itself fixed by the CFT [19], so they can be included explicitly in the fit and c extracted absolutely, without a reference. The costs differ: the absolute route needs a window in T long enough to separate the correction from the leading term, while calibration works at shorter times but needs the reference simulation.

C Truncation schemes

For the general framework of tensor networks—diagrammatic notation, matrix product states and operators, canonical forms and the standard algorithms—we refer to the reviews of Refs. [6, 7]. This appendix records only the linear algebra behind the truncation choice of Section 3.4: why truncating on Schmidt values is optimal for a single state, why gauge freedom decides which spectra are physical, and how the construction generalizes to the **RTM**.

C.1 Gauge freedom and canonical forms

The **MPS** representation of a state is not unique. On any virtual bond one may insert $\mathbb{1} = XX^{-1}$ and absorb the two factors into the neighbouring tensors, $A^{[n]} \rightarrow A^{[n]}X$ and $A^{[n+1]} \rightarrow X^{-1}A^{[n+1]}$, changing the tensors while leaving the state untouched. The canonical form fixes this freedom: sweeping QR or SVD decompositions inward from both ends makes every tensor an isometry, $A^\dagger A = \mathbb{1}$ from the left and $BB^\dagger = \mathbb{1}$ from the right, with one centre tensor carrying the non-trivial weight. Refs. [6, 7] give the construction in full.

Gauge freedom draws the line between physical and representational quantities. Anything that depends on the individual tensors rather than on the state they encode can be changed at will by a gauge transformation, and is therefore not meaningful to truncate on. In the single-state setting this causes no trouble, because the canonical form fixes the gauge and the Schmidt values below are automatically the physical ones. In the transverse picture it becomes the deciding issue, as Section C.3 shows.

C.2 Schmidt values and optimal truncation

Consider a cut dividing the chain into a block $\mathcal{A}$ and its complement $\mathcal{B}$. Grouping the physical indices on each side into one composite index turns the wavefunction into a matrix, whose Singular Value Decomposition (**SVD**) gives the Schmidt decomposition

$$|\psi\rangle = \sum_{\alpha=1}^{\chi} s_{\alpha} |\alpha\rangle_{\mathcal{A}} \otimes |\alpha\rangle_{\mathcal{B}}, \quad s_{\alpha} \geq 0, \quad \sum_{\alpha} s_{\alpha}^2 = 1, \quad (33)$$

with orthonormal bases on the two sides. In mixed canonical form this structure sits on the cut bond: the centre object is the diagonal matrix of Schmidt coefficients, and the **RDM** of the block is diagonal in the same basis with eigenvalues $p_{\alpha} = s_{\alpha}^2$, the probabilities that fix the entanglement across the cut.

Truncating a bond from χ to $\chi' < \chi$ means keeping the χ' largest Schmidt coefficients. This is optimal in a precise sense: by the Eckart–Young–Mirsky theorem, among all states of Schmidt rank at most χ' the truncated state is the closest to $|\psi\rangle$ in norm, with an error equal to the discarded weight,

$$\varepsilon = \|\psi\rangle - |\tilde{\psi}\rangle\|_2^2 = \sum_{\alpha > \chi'} s_{\alpha}^2. \quad (34)$$

Figure 11 shows why the truncation can be performed locally in canonical form.

Two features of this scheme matter later, because the transverse picture keeps one and loses the other. First, the truncated object is Hermitian and positive, so its eigenvalues are the squared Schmidt values and dropping the smallest is unambiguous. Second, the Eckart–Young–Mirsky theorem provides a variational guarantee and bounds the error. Both are properties of the Hermitian, single-state setting, and neither survives the move to a non-Hermitian transition matrix.

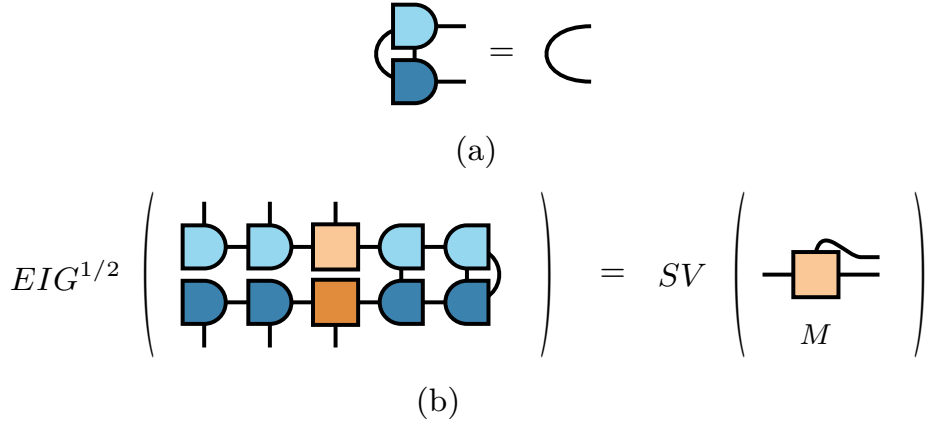


Figure 11: **Why the RDM truncation is local.** (a) The canonical condition: a tensor contracted with its own conjugate over the physical index and one virtual index reduces to the identity on the remaining one. A tensor and its conjugate are drawn as the same shape in a lighter and a darker shade. (b) The reduced density matrix at a cut, with the left block open and the right block traced. In mixed canonical form the isometries either side of the orthogonality centre collapse by (a), so the eigenvalues of the whole object are the squared singular values of the single tensor M , and the optimal truncation never touches the rest of the chain.

C.3 The reduced transition matrix and its truncation

In the transverse picture the transfer matrix is non-Hermitian, so its dominant left and right fixed points are distinct states, and the object to truncate is the **RTM** of Eq. (11), built from both. Its eigenvalues can be complex, so the Schmidt interpretation above does not carry over. The gauge freedom also controls how well the two fixed points can be represented at a given bond dimension, since it moves weight between the components that contribute to $\langle L|R \rangle$ and those that do not [42]. Figure 12 contrasts the reduced density matrix with the transition matrix at the same temporal cut.

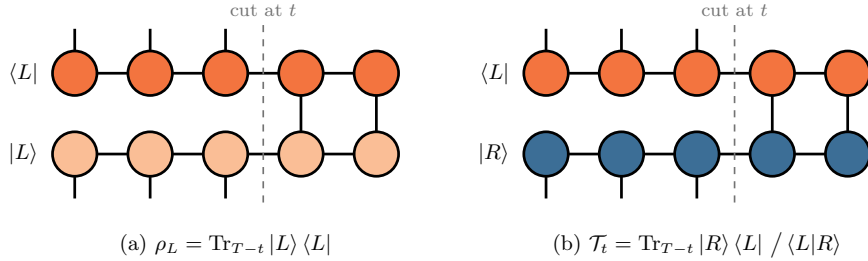


Figure 12: **The two candidate objects at a temporal cut.** Sites after the cut are traced over, so their physical legs are contracted between the two rows; the legs before the cut stay open and carry the indices of the resulting matrix. (a) The reduced density matrix of a single temporal boundary pairs $|L\rangle$ with its own conjugate, and is Hermitian and positive. (b) The **RTM** instead pairs the two distinct fixed points $|R\rangle$ and $\langle L|$, so it is not Hermitian and its eigenvalues need not be real or positive.

The choice between the two objects is settled by their behaviour under a gauge transformation. Insert X on the cut bond: the reduced density matrix and the transition matrix respond differently. The boundary RDM transforms by a congruence, $\rho_L \rightarrow X^\dagger \rho_L X$, which for the non-unitary X generic here distorts its spectrum. The transition matrix transforms by a similarity, $\mathcal{T}_t \rightarrow X^{-1} \mathcal{T}_t X$, which leaves its eigenvalues untouched. Fig-

ure 13 shows the difference at the level of the network. The Schmidt values of a single temporal boundary are therefore not physical, while the RTM spectrum is, and truncating on the former can discard exactly the weight the overlap needs.

The gauge argument does not make the RDM scheme unusable. As long as one works in a fixed canonical form the gauge is pinned, and the RDM truncation is well defined and behaves exactly as it does in any ordinary tensor-network calculation. What the argument establishes is which of the two spectra is a property of the physics rather than of the representation, and therefore which one the CFT constrains.

A second question is why the RTM scheme truncates on singular values at all, when the RDM scheme truncates on eigenvalues. Neither scheme ever diagonalises the exponentially large global object. Canonical forms reduce both to a local $\chi \times \chi$ environment matrix M_A sitting on the cut bond, and the two reductions differ. For the Hermitian case a bipartite cut gives

$$\rho_B = U_B M_A U_B^\dagger, \quad (35)$$

with U_B unitary because it is built from left-orthogonal tensors. Equation (35) is a similarity transformation, so $\text{eig}(\rho_B) = \text{eig}(M_A)$: the eigenvalues of the small environment are the eigenvalues of the full reduced density matrix, and one may truncate on them rigorously. For the RTM, built from two different states, the same cut instead gives

$$\tau_B = U_B M_A V_B, \quad U_B \neq V_B^\dagger, \quad (36)$$

which is no longer a similarity transformation, so the eigenvalues are not preserved and $\text{eig}(\tau_B) \neq \text{eig}(M_A)$. Fortunately U_B and V_B are still isometries, and isometries preserve singular values,

$$\text{sv}(\tau_B) = \text{sv}(M_A). \quad (37)$$

Equation (37) is what makes the scheme possible: the singular values of the small environment are the singular values of the full object, so they are the quantity we are entitled to truncate on. Using them is not a modelling choice but the only invariant the reduction leaves us.

This last step requires care, though, because singular values and eigenvalues answer different questions. They are related by $\text{sv}(M) = \sqrt{\text{eig}(MM^\dagger)}$, and for a Hermitian positive RDM the two coincide, so discarding the smallest singular values is the same as discarding the smallest eigenvalues. For the non-Hermitian RTM they do not, and the quantity we actually need is the trace $\text{Tr } \mathcal{T}_t = \sum_i \lambda_i$, which gives the physical overlap. A trace is a signed sum, so it is not controlled by singular-value magnitudes. The simplest illustration is an RTM with the two eigenvalues $+1$ and -1 : both have modulus one, so an SVD would rank them among the most important components and keep them, yet they cancel exactly in the trace and contribute nothing to the physical answer.

Truncating the RTM therefore lacks the rigorous bound that the RDM enjoys, and it would very likely fail on a generic random non-Hermitian matrix. The reason it nevertheless works here is physical rather than mathematical: the transfer matrices generated by a time evolution are highly structured, not generic, and in practice the cancellations that would defeat the criterion do not occur. This is an empirical statement about the operators we meet, not a theorem, which is why we keep the RDM scheme available as an independent check.

The canonical form still helps, but in a generalized version, shown in Figure 14: the condition of Figure 11(a) is imposed on the pair $\langle L|, |R\rangle$ rather than on a state and its own conjugate. The tensors away from the cut then contract away exactly as before, and

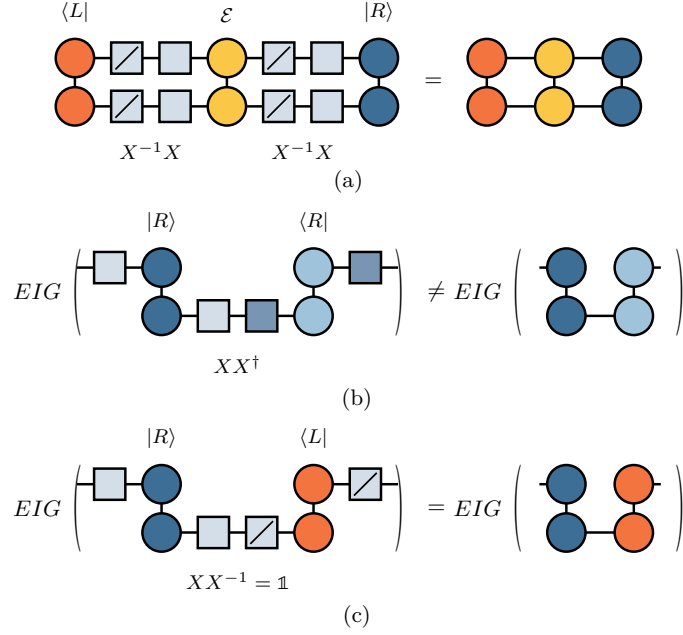


Figure 13: **Why the RTM spectrum is the gauge-invariant one.** (a) A pair $X^{-1}X$ may be inserted on every bond at no cost: the two factors cancel, so the contraction equals the ungauged one and the Loschmidt echo is untouched, while the individual tensors change. In (b) and (c) only the lower time site is contracted; the upper one carries the open indices, so each open index keeps one factor and the contracted bond receives both. (b) The **RDM** pairs $|R\rangle$ with its own conjugate, the factors meet as $XX^\dagger \neq \mathbb{1}$, and the spectrum moves under the congruence $X\rho X^\dagger$. (c) The **RTM** pairs $|R\rangle$ with $\langle L|$, the factors meet as $XX^{-1} = \mathbb{1}$, and the similarity $X\mathcal{T}_t X^{-1}$ leaves the spectrum unchanged.

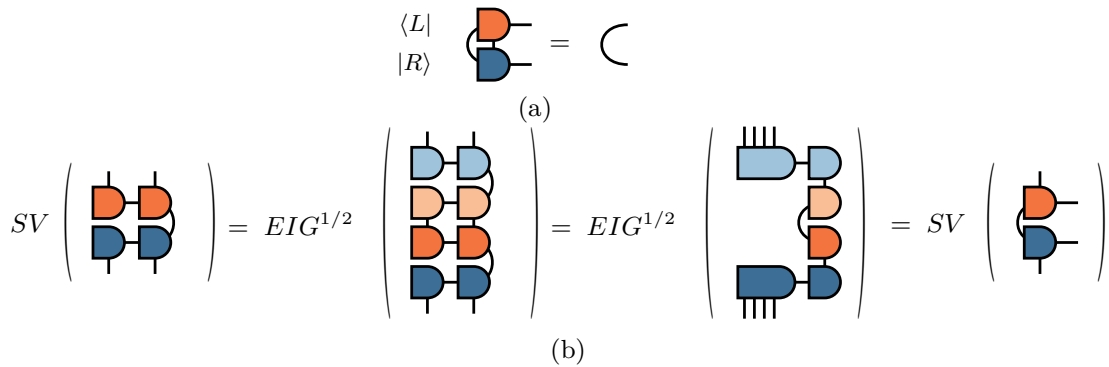


Figure 14: **The RTM counterpart of Figure 11.** (a) The generalized canonical condition: the contraction pairs a tensor of $\langle L|$ with the corresponding tensor of $|R\rangle$, not with its own conjugate. (b) Imposed along the chain, this condition makes everything before the cut contract away, so the singular values of the **RTM** are those of the two tensors at the cut alone. The construction is as local as the **RDM** one; what it loses is the variational guarantee, since the object is no longer Hermitian.

the singular values of the **RTM** again follow from a local decomposition, so the two routes cost the same to evaluate and differ only in which object they compress.

The logic of truncation changes accordingly. In the ordinary case one keeps the directions that best preserve the norm of a state. Here the relevant quantities are bilinear matrix elements $\langle L | \mathcal{O} | R \rangle$. A component of $|R\rangle$ with little overlap with $\langle L|$ contributes little, however large it looks from $|R\rangle$ alone, and a component with small weight in $|R\rangle$ can still matter if its projection onto $\langle L|$ is large. The scheme used here therefore performs a biorthogonal decomposition of $|R\rangle \langle L|$ and keeps the eigenvalues of largest modulus [27], which are the directions that dominate the physical contraction. Because relevance is measured through the overlap, this can retain fewer states than an RDM truncation at the same accuracy.

The stability caveat of Section 3.4 can now be made precise. The singular values of $\mathcal{T}_t$ and its eigenvalues are different quantities, and it is the eigenvalues that enter the contraction, since the trace is their sum. A component with a large singular value can therefore still fail to contribute: an operator with eigenvalues $+1$ and -1 has both at unit modulus, so an SVD-based truncation keeps both, yet they cancel exactly in the trace. Truncating the RTM on singular values is thus a motivated heuristic rather than a bound.

A second approximation enters through the environment. The truncation at each step uses the local environment available at that point of the sweep, not the full remaining network, because the contraction is built column by column. For a translation-invariant problem this is benign, since every bulk column is the same operator and weight discarded too early is regenerated by the next application. It remains an approximation, and it is one reason the RDM route behaves more predictably when the RTM output turns noisy.

D The fermionic form of the NNN interactions

This appendix derives the fermionic form of the model: the **NNN** coupling becomes a genuine interaction after fermionization, which is what breaks integrability for every $p \neq 0$.

The effect of the additional interactions becomes explicit after mapping the spin chain to fermions. In the convention of Eq. (13), with the transverse field along σ^x and the order parameter along σ^z , the **JW** transformation [43, 44] is

$$\sigma_j^x = 1 - 2n_j, \quad \sigma_j^z = (c_j + c_j^\dagger) K_j, \quad K_j = \prod_{k < j} (1 - 2n_k), \quad (38)$$

where c_j annihilates a spinless fermion, $n_j = c_j^\dagger c_j$, and the string K_j enforces the fermionic anticommutation relations.

The two Ising terms remain quadratic under this transformation. The transverse field becomes a chemical-potential term, while the strings associated with the **NN** bond cancel and leave ordinary hopping and pairing. The $p = 0$ chain is therefore a free-fermion model and can be diagonalised exactly [35]. At criticality, its continuum limit is the free Majorana **CFT** with $c = \frac{1}{2}$ [23].

The terms proportional to p introduce quartic fermionic contributions. The self-dual $\sigma_j^x \sigma_{j+1}^x$ term becomes the density–density factor $(1 - 2n_j)(1 - 2n_{j+1})$, which is diagonal in the occupation basis. For the **NNN** bond, the Jordan–Wigner strings do not cancel completely: a parity factor remains on the intermediate site,

$$\sigma_i^z \sigma_{i+2}^z = (c_i^\dagger - c_i) (1 - 2n_{i+1}) (c_{i+2} + c_{i+2}^\dagger). \quad (39)$$

This term describes hopping and pairing between sites i and $i+2$, with an amplitude whose sign depends on the occupation of site $i+1$. We refer to this process as parity-dressed

hopping. The quartic contributions allow the fermions to scatter, so Wick’s theorem no longer reduces correlation functions to two-point functions [21]. Consequently, the model is genuinely interacting and no longer admits a free-fermion solution for $p \neq 0$. The algebra behind these statements is given below.

D.1 Jordan–Wigner and the four-fermion interaction

The main text quotes the transformation in Eq. (38) and its action on the NNN bond in Eq. (39); this subsection supplies the algebra between the two. The string is what makes the transformation possible at all. Two Pauli operators anticommute on the same site but commute on different ones, so a spin is neither a boson nor a fermion, and no purely local combination of spins can anticommute at a distance. Attaching to each fermion the parity of everything to its left converts the commuting spin algebra into the anticommuting fermion one exactly. The reward is that a Hamiltonian quadratic in fermions is exactly solvable — diagonalizing a single-particle matrix yields independent normal modes — which is the reason the plain Ising chain is free and its critical point a free Majorana CFT. The working identities below are $(c + c^\dagger)(1 - 2n) = c^\dagger - c$ and $K_j^2 = 1$.

The transverse field becomes a chemical potential directly from Eq. (38). For the NN Ising bond the two strings overlap on all but one site, so they cancel telescopically and leave the quadratic hopping-and-pairing term

$$\sigma_j^z \sigma_{j+1}^z = (c_j^\dagger - c_j)(c_{j+1} + c_{j+1}^\dagger). \quad (\text{F1})$$

These two terms make up the pure Ising chain, which is why it is a free-fermion theory, exactly solvable, with a free Majorana CFT at criticality.

The two terms proportional to p are both quartic. The self-dual $\sigma^x \sigma^x$ term becomes the density–density factor $(1 - 2n_j)(1 - 2n_{j+1})$, diagonal in the occupation basis. For the NNN bond the strings cancel only partially: the site skipped over contributes its factor once instead of twice, leaving the parity $(1 - 2n_{i+1}) = (-1)^{n_{i+1}}$ in place, which is Eq. (39)—a hop from i to $i + 2$ whose sign is set by the occupation of the intermediate site.

E Extraction and equilibrium details

This appendix collects the equilibrium measurements and fit conventions behind numbers quoted in the main text: the finite-size check of the universality class, the sound velocity, and the branch constant of the eigenvalue fit.

E.1 Finite-size scaling of the mid-chain entropy

Section 4.1 reads the central charge from the shape of the entanglement profile at a single system size. A second, independent estimate comes from the size dependence of one point of that profile. At the mid-chain cut the chord takes its maximum, $W(L/2, L) = \ln(2L/\pi)$, so Eq. (14) collapses to

$$S_{\text{vN}}(L/2) = \frac{c}{6} \ln L + k'. \quad (\text{G0})$$

Running DMRG over a ladder of sizes and plotting the mid-chain entropy against $\ln L$ therefore gives a straight line of slope $c/6$. Over $N = 40$ to 200 this returns $c \approx 0.51$ at both $p = 0.1$ and $p = 0.5$, consistent with the profile fit of the main text and with the Ising value. The ladder uses fifteen DMRG sweeps with the bond dimension ramped to

400 and a truncation cutoff of 10^{-10} ; the heavier $N = 400$ chord fit of Section 4.1 uses fifty sweeps, a ramp to 1000, and a cutoff of 10^{-12} . Both ramps start with noise-assisted sweeps.

The same ground states allow a cross-check with the Rényi-2 entropy, the member used for every temporal measurement in this thesis. Its mid-chain slope is $c/8$ rather than $c/6$, and a plain linear fit returns $c \approx 0.58$: the same universality class, but biased upward. The bias is expected. Rényi entropies with $n > 1$ carry corrections to scaling that the von Neumann entropy does not, oscillating with the cut position and decaying as a power of the subsystem size [34], so a straight-line fit that ignores them overestimates the slope. The temporal analogue of this bias is the finite-time correction $\Delta S_n \propto T^{-1/n}$ [19], which decays more slowly as n grows; it is fitted where it matters, in Section 7.1. We do not fit the equilibrium corrections here, because the von Neumann route already measures the equilibrium central charge cleanly, which is why it is the estimator of choice where the full Schmidt spectrum is available. Figure 15 compares the two finite-size estimates.

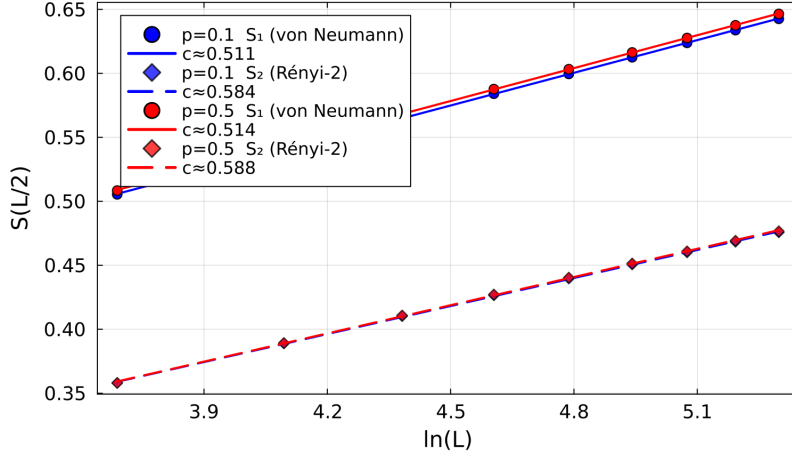


Figure 15: **Finite-size scaling of the mid-chain entropy.** At $p = 0.1$ and $p = 0.5$; the von Neumann entropy (solid, slope $c/6$) gives $c \approx 0.51$; the Rényi-2 entropy (dashed, slope $c/8$), computed from the same ground states, gives $c \approx 0.58$, with the larger corrections to scaling characteristic of $n > 1$.

E.2 The sound velocity from the equilibrium dispersion

Section 4.2 measures the sound velocity from the equilibrium dispersion. This appendix gives the construction in full and records the checks behind the numbers quoted there.

The measurement uses a periodic chain, so that momentum is a good quantum number. We build the ring Hamiltonian of Eq. (13) at $\lambda = 1$ for a small number of sites and diagonalize it exactly. Because the ring is translation invariant, every eigenstate can be labelled by its momentum, which we read off from the expectation value of the one-site translation operator, and by its parity under the spin flip. Degenerate levels come out of the diagonalization as arbitrary mixtures, so we resolve momentum and parity together before assigning labels. Within the magnetic (odd-parity) sector we then take the lowest state carrying no momentum and the lowest state carrying momentum k , and the velocity follows from the slope of the dispersion between them,

$$v = \frac{N}{2\pi} \frac{E_1(k) - E_1(0)}{|k|}, \quad (40)$$

with k the smallest non-zero momentum available on the ring. Equation (40) is the lattice form of the linear dispersion $\omega = v|k|$: on a ring of N sites the allowed momenta are spaced by $2\pi/N$, so the factor $N/2\pi$ converts an energy difference between neighbouring momentum levels into the slope of that line. Everything is computed at $\lambda = 1$, where the chain is critical and the dispersion is linear at long wavelengths.

The construction is validated in four ways. It recovers $v = 2$ at $p = 0$, to 1.990 at $N = 18$ and improving with size. The ring Hamiltonian built from scratch matches the operator used elsewhere in the code to machine precision, 7×10^{-14} on the full spectrum at $N = 8$. The sector-resolved and sector-blind reads agree at every p . Finally, feeding the measured $v(p)$ into Alcaraz’s separate Casimir formula reproduces his published ground-state central charge: we obtain $c = 0.504$ at $p = 0$ and $c = 0.525$ at $p = 0.5$ from rings of 16 sites.

The ring sizes that exact diagonalization reaches are not in the thermodynamic limit. At $p = 0$ the measurement still returns 1.9899 at $N = 18$, against the exact 2. That error is not random: multiplying it by N^2 gives 3.27, 3.28, 3.28, 3.28 and 3.28 at $N = 10$ to 18, so it decays as $1/N^2$. Reaching three-digit accuracy by enlarging the ring would need $N \approx 57$, far beyond what can be diagonalized, but the regularity of the error means it can be fitted away instead.

We therefore fit $v(N) = v_\infty + C/N^2$ at each coupling and take v_∞ as the production velocity. Table 3 collects the measurements and the fits, and Figure 16 shows them. The residuals stay below 2.8×10^{-2} at every coupling, on velocities of order two to eleven, so the assumed form describes the data rather than being imposed on it. The $p = 0$ row is the control: the extrapolation returns 1.99994 against the exact 2, an error of 6×10^{-5} where the largest ring is still 1.0×10^{-2} away. The correction grows steeply with frustration, from $C = -3.27$ at $p = 0$ to -140.7 at $p = 1.5$, so the extrapolation matters most at the couplings where the raw ring values sit furthest from the limit. It is these extrapolated values, rather than a fixed $v = 2$, that make the measured boundary dimensions read as clean half-integers at every p .

p	$N = 10$	$N = 12$	$N = 14$	$N = 16$	$N = 18$	v_∞	C
0.0	1.9673	1.9772	1.9833	1.9872	1.9899	1.99994	-3.27
0.1	2.5707	2.6007	2.6191	2.6312	2.6396	2.67016	-9.97
0.2	3.1565	3.2073	3.2387	3.2594	3.2738	3.32561	-16.96
0.3	3.7247	3.7972	3.8423	3.8722	3.8931	3.96723	-24.34
0.4	4.2756	4.3707	4.4305	4.4703	4.4981	4.59585	-32.16
0.5	4.8096	4.9284	5.0038	5.0543	5.0897	5.21214	-40.46
1.0	7.2534	7.5020	7.6674	7.7824	7.8653	8.12650	-88.23
1.5	9.4140	9.7946	10.0594	10.2504	10.3921	10.79924	-140.69

Table 3: **The measured velocities and their extrapolation.** $v(p)$ at $\lambda = 1$ from rings of $N = 10$ to 18 sites, with the parameters of the fit $v(N) = v_\infty + C/N^2$. The largest fit residual is 1.1×10^{-5} at $p = 0$ and 2.8×10^{-2} at $p = 1.5$. The v_∞ column holds the velocities used throughout this work.

Over the range studied the extrapolated velocity is close to linear in p : fitting $v_\infty = K_0 + K_1 p$ gives $2.02 + 6.42 p$, with residuals up to 0.023. This is a description of the data and not a law. The same fit applied to the individual ring sizes has coefficients that still move with N , from $(K_0, K_1) = (2.00, 5.68)$ at $N = 10$ to $(2.01, 6.13)$ at $N = 16$, and the intercept of the extrapolated line misses the exact $v(0) = 2$ by more than the extrapolation itself does. At stronger frustration than we reach here the linear description would be expected to fail altogether. Every conversion in this thesis therefore uses v_∞ at

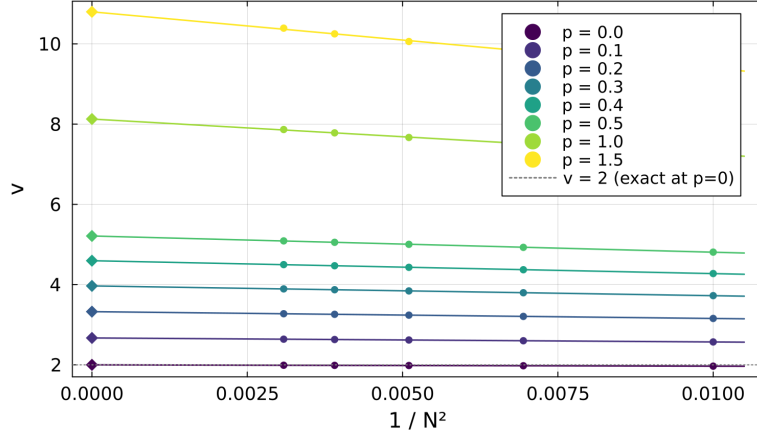


Figure 16: **Extrapolating the velocity to the thermodynamic limit.** The measured v against $1/N^2$ at each coupling, with the fitted lines and their intercepts (diamonds). The intercept is the production velocity. At $p = 0$ it lands on the exact $v = 2$, marked by the dotted line, which is the control on the procedure.

its own coupling, never the fitted line.

That the temporal transfer matrix really carries this p -dependent velocity was checked separately with the anisotropic XY control, whose velocity $v(\gamma) = 2\gamma$ is known exactly. At $\gamma = 0.5, 1, 1.5$ the measured first-gap slopes are $a_1 = 1.481, 0.782, 0.520$ against the exact $\pi/(4\gamma) = 1.571, 0.785, 0.524$, giving $x_1 = 0.471, 0.498, 0.496$; pinning $v = 2$ instead would force $a_1 = 0.785$ at every γ and disagree with two of the three measured slopes. The pipeline recovers $x_1 = \frac{1}{2}$ only when fed the correct $v(\gamma)$, which confirms that the method is velocity-faithful.

E.3 Fit windows and the controls behind the quoted numbers

The central charges of Table 2 come from least-squares fits, and two conventions fix them: which times enter the fit, and what is held fixed. This appendix records both, together with the checks that show how far each number can move.

The window has no unique start, and the choice matters. Moving the first fitted time later shortens the ladder and pushes the extrapolation upwards, because the $1/T$ and $1/T^2$ terms of Eq. (6) become degenerate over a short range. Scanning the window start moves the result by 0.014 at $p = 0$, 0.024 at $p = 0.1$ and 0.21 at $p = 0.3$. Every value quoted is taken from the widest window that passes the consistency test below, because that window has the most points and the least degeneracy between the two correction terms. The window ends where the phase of μ_0 stops advancing at a constant rate: over $T = 2.2$ to 4.5 at $p = 0.5$ that rate is 1.8326 with a spread of 0.0008, and beyond it the rate scatters. That criterion is sharper than any test on $|\mu_0|$ or on the rigidity, and it is what sets the windows in Table 2.

The branch constant of Eq. (6) is pinned rather than fitted. Its value is known from the definition $\lambda_0 = \log(-\mu_0)$, and where the window is long enough to determine it independently the fit returns $B/\pi = -1.000$, so pinning costs nothing and removes a parameter that a short window cannot constrain. The phase itself is unwrapped over the whole ladder before any window is applied: unwrapping inside a window discards the absolute branch, and the pinned fit is then meaningless.

Three controls bound how far the numbers can move. Repeating a ladder at $\chi = 128$ changes the imaginary plateau by about 2% and $|\mu_0|$ by less than 10^{-4} , while the chord slope of the real part moves by up to 0.12; this is why the entropy measurements are read from the plateau and not from the slope. Halving the Trotter step, with N_β doubled so that β_0 is unchanged, moves the plateau by under 1% well inside the window and by 18% at its edge. Repeating a single rung with five independent random seeds spreads the plateau by 3% and the chord slope by 12%, while $|\mu_0|$ agrees to 1.3×10^{-4} , so the spectral route is far less exposed to the seed than the entropy route. The sound velocity enters linearly, and the extrapolation that produces it returns 1.99994 against the exact 2 at the integrable point.

For the absolute entropy result of Eq. (21) the correction law is a further choice, since it is not known in advance. The plateau is fitted with a constant, with $a + bT^{-1/2}$, with $a + b/T$ and with $a + b/T + c/T^2$, and the four are compared by corrected Akaike information. Over $T = 2$ to 24 the constant is rejected by a difference of 21, while the $T^{-1/2}$ and two-term forms differ by less than 1 and give 0.506 and 0.511. Over the short window $T \leq 11$ the same comparison cannot separate any of the four, which is why the long ladder is necessary.

F Construction of the time-evolution MPO

This appendix records the construction details behind Section 5: how the exponentiation is carried out in practice, the two schemes it can apply, the closed-form second-order tensor used in production, and the measurement that justifies it.

F.1 How the exponentiation is carried out

The exponentiation is performed with the `ITensorExpMPO.jl` package [45], which implements the constructions of Zaletel et al. [41] on top of `ITensors.jl` [46]. The package takes the Hamiltonian in the symbolic form used to build any MPO in that library, a sum of operator strings with their coefficients, so the FSM of Section 5.1 never has to be written out by hand. It walks along the chain, collects at every bond the terms that cross it, and compresses the resulting coefficient matrix with an SVD. What survives is the smallest set of memory channels able to carry the interactions across that bond, which for Eq. (13) is the three channels of Eq. (15). The blocks A , B , C and D are then read off site by site and passed to the routine that builds the local tensor of the evolution operator.

That routine is where our own contribution sits. The package provided the first- and second-order operators of Zaletel et al.; we added a mode that applies the same machinery with the second-order recipe of Van Damme et al. [40], together with the enlargement of the virtual bond from $1 + \chi$ to $1 + \chi + \chi^2$ that this recipe requires. The addition is available in a fork of the package [47], and the two repositories document the software details.

F.2 The two constructions in outline

Both constructions solve the same problem. Reaching second order by squaring W_H directly produces cross terms between the blocks, since H^2 sums over every ordered pair: both orderings CD and DC of an initiator and a field on the same site must be kept, or the operator loses the Hermiticity it inherits from H . Writing out the correct symmetric combinations by hand is impractical. The first construction avoids this by encoding the blocks in an auxiliary space of two hard-core bosons, where the exponential terminates by itself once both modes are filled; exponentiating that small operator and projecting onto

the four boson states returns the blocks of W directly [41]. Keeping only the terms with disjoint support gives W^I , which is first order, and adding the single-site overlaps gives W^{II} , which is second order provided the Hamiltonian has only **NN** interactions. Both keep the virtual bond at $1 + \chi$.

The second construction organizes the same expansion by clusters of overlapping terms rather than by powers of W_H [40]. Terms with disjoint support commute and factorize, so they are kept to all orders, and only the connected clusters have to be reproduced explicitly; retaining clusters up to size n leaves a one-step error of order δt^{n+1} . The second-order member is the operator used here. It carries a doubled memory leg, which is the origin of the virtual bond $1 + \chi + \chi^2$, and that leg is what reproduces the three-site clusters generated by the **NNN** bond. The original papers give the derivations; the resulting tensor is recorded below.

All of these schemes share one device, which is what keeps them stable. The complex step $\tau = -i\delta t$ is split unevenly between the blocks: the initiator C carries a real factor $t_C = \sqrt{\delta t}$ and the terminator B the phase $t_B = \tau/t_C$, so the product $C \otimes B$ still carries the correct factor while the intermediate memory channels stay well behaved.

F.3 The second-order $VD2$ blocks

We record the explicit local tensor of the second-order MPO ($VD2$) used throughout this work. These are the closed-form expressions of Van Damme et al. [40], specialized to our **FSM** blocks; they are what the exponentiation routine assembles for a given memory dimension χ .

We start from the four blocks A, B, C, D of the Hamiltonian MPO W_H (Eq. (15)): D is the on-site term, C initiates an interaction, B terminates one, and A propagates it. Following the time-step split above, we scale them as

$$\tilde{D} = \tau D, \quad \tilde{C} = t_C C, \quad \tilde{B} = t_B B, \quad \tilde{A} = A, \quad t_C = \sqrt{\delta t}, \quad t_B = \tau/t_C, \quad \tau = -i\delta t. \quad (41)$$

The construction organizes the exponential $e^{\tau W_H}$ not by powers of W_H , but by the symmetric products of overlapping blocks that its Taylor series generates. We write $\{X_1 \dots X_m\}$ for the symmetrized product, i.e. the sum over all distinct orderings of the operator-valued blocks (for example $\{XY\} = X \otimes Y + Y \otimes X$, and $\{XXY\} = XXY + XYX + YXX$), where $\otimes$ is the Kronecker product on the virtual memory index and the ordinary product on the physical index. The multinomial weights $1/2!, 1/3!, \dots$ of the exponential are carried by the explicit prefactors below.

The resulting local tensor is a 3×3 block object (indices 1, 2, 3 label the scalar, single- and doubled-memory sectors, which is the origin of the enlarged $1 + \chi + \chi^2$ virtual bond),

$$W^{VD2} = \begin{pmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \\ W_{31} & W_{32} & W_{33} \end{pmatrix}, \quad (42)$$

with

$$W_{11} = \mathbf{1} + \tilde{D} + \frac{1}{2}\{\tilde{D}\tilde{D}\} + \frac{1}{6}\{\tilde{D}\tilde{D}\tilde{D}\}, \quad (43)$$

$$W_{12} = \tilde{C} + \frac{1}{2}\{\tilde{C}\tilde{D}\} + \frac{1}{6}\{\tilde{C}\tilde{D}\tilde{D}\}, \quad W_{13} = \{\tilde{C}\tilde{C}\} + \frac{1}{3}\{\tilde{C}\tilde{C}\tilde{D}\}, \quad (44)$$

$$W_{21} = \tilde{B} + \frac{1}{2}\{\tilde{B}\tilde{D}\} + \frac{1}{6}\{\tilde{B}\tilde{D}\tilde{D}\}, \quad W_{31} = \frac{1}{2}\{\tilde{B}\tilde{B}\} + \frac{1}{6}\{\tilde{B}\tilde{B}\tilde{D}\}, \quad (45)$$

$$W_{22} = \tilde{A} + \frac{1}{2}(\{\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{D}\}) + \frac{1}{6}(\{\tilde{C}\tilde{B}\tilde{D}\} + \{\tilde{A}\tilde{D}\tilde{D}\}), \quad (46)$$

$$W_{23} = \{\tilde{A}\tilde{C}\} + \frac{1}{3}(\{\tilde{A}\tilde{C}\tilde{D}\} + \{\tilde{C}\tilde{C}\tilde{B}\}), \quad W_{32} = \frac{1}{2}\{\tilde{A}\tilde{B}\} + \frac{1}{6}(\{\tilde{A}\tilde{B}\tilde{D}\} + \{\tilde{B}\tilde{B}\tilde{C}\}), \quad (47)$$

$$W_{33} = \{\tilde{A}\tilde{A}\} + \frac{1}{3}(\{\tilde{A}\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{A}\tilde{D}\}). \quad (48)$$

The scalar track W_{11} carries the on-site evolution to third Taylor order; the W_{12}, W_{13} entries dress an outgoing memory leg (single and doubled), W_{21}, W_{31} the incoming one, and the lower-right block dresses the through-wire $\tilde{A}$ with all of its symmetric companions. Keeping the doubled-leg (index-3) sector is exactly what reproduces the connected clusters spanning three sites—the **NNN** overlaps that W^{II} misses—and is why *VD2* is genuinely second order for our model at the cost of the enlarged virtual bond.

F.4 Comparing the two constructions in practice

The claim that W^{II} degrades to first order for a Hamiltonian with **NNN** interactions, while *VD2* remains second order, can be checked directly on the model of this thesis. A convenient diagnostic is the loss of unitarity, since the exact $e^{-iH\delta t}$ preserves the norm of any state: applying successive layers of the approximate operator to $|X+\rangle^{\otimes N}$ and monitoring $\|\psi\|$ isolates the Trotter error, because a perfect construction would leave the norm at one.

Applying eight layers at $\delta t = 0.05$ to a chain of $N = 100$ sites at $p = 0.1$ gives the following norms:

	W^{I}	W^{II}	<i>VD2</i>
virtual bond	4	4	13
$\ \psi\ $ after 1 layer	1.3514	1.1341	0.99995
$\ \psi\ $ after 8 layers	12.6146	3.0181	0.99923

The separation is unambiguous. The first-order W^{I} inflates the norm by an order of magnitude within eight steps, and W^{II} , which would be effectively second order for a strictly **NN** chain, still triples it: the connected clusters generated by the **NNN** bridge are precisely the ones it fails to capture. *VD2* holds the norm to within 8×10^{-4} over the same interval.

This is what justifies paying for the larger virtual bond. Since the transverse contraction needs long temporal chains, and reducing δt to compensate a first-order error would lengthen them further and add truncation error at every extra site, a genuinely second-order operator at a moderate time step is the configuration that makes the calculation feasible at all.

G Implementation of the block power method

This appendix documents the implementation of the block power method of Section 6.1. The core routine is `block_transfer_eigs`, built on top of `ITransverse.jl` [28]; all code, notebooks and cluster drivers are in the public repository for this thesis [48].

G.1 The regularized solve and the left pairing

Near a degeneracy the columns of the right block begin to coincide and the metric S of Eq. (19) becomes nearly singular. A generalized eigensolver applied to the pencil $Mv = \theta Sv$ would then return infinite or scattered values, so we regularize it with the pseudoinverse instead,

$$W = S^+ M, \quad W = V \Theta V^{-1}, \quad (\text{D1})$$

where S^+ carries a relative tolerance of 10^{-12} : directions that have effectively collapsed onto each other are projected out rather than inverted. The eigenvalues Θ , sorted by modulus, estimate $\mu_0, \dots, \mu_{k-1}$, and the columns of V give the right mixing coefficients.

The left block needs its own coefficients, and obtaining them from a second diagonalization is unreliable: the transposed pencil returns its eigenvalues in no guaranteed order, and inside a quasi-degenerate cluster the matching of left to right eigenvalues becomes ambiguous exactly when it matters, so the paired vectors can stop describing the same eigenvector and the block oscillates instead of converging. Fortunately, no second diagonalization is needed. Writing $y^T = u^T S$, the left problem becomes $y^T W = \theta y^T$, so the y are the rows of V^{-1} and

$$u_j = (S^+)^T (V^{-1})^T e_j, \quad (\text{D2})$$

paired with θ_j by construction and biorthogonal through the metric, $u_i^T S v_j = \delta_{ij}$.

One implementation note concerns the de-mixing. The MPS linear combinations of Section 6.1 are formed by exact direct-sum addition and compressed once at the end, because the default density-matrix addition is fragile for the near-parallel vectors that a tight cluster produces.

G.2 The eigenvalue-only mode

The eigenvalue-only mode of Section 6.1 replaces the de-mixing coefficients by orthonormal ones. After the modulus sorting, the coefficient matrices are QR-decomposed,

$$V \rightarrow Q_V, \quad U \rightarrow Q_U, \quad Q^\dagger Q = \mathbb{1}, \quad (\text{D3})$$

and the blocks are rotated with Q_V and Q_U instead of V and U . The leading j columns of each Q span the same leading subspaces as those of the sorted coefficient matrices, so this is an ordered-Schur-like choice. The Ritz values are unchanged because recombining the blocks by any invertible B and C sends $S \rightarrow B^T S C$ and $M \rightarrow B^T M C$, which leaves the solutions of the projected pencil untouched; the two bases differ only in how well conditioned they are when truncated. What the rotation does change is the conditioning of what is truncated next: $\text{cond}(V)$ grows like the inverse of the gap inside the block, while Q_V and Q_U are unitary by construction. The two modes can therefore drift apart only through the truncation, and in practice they barely do. Two independent ladders at $p = 0.1$ and $\chi = 64$, one carried in each basis, agree on the two tracked eigenvalues to between 10^{-7} and 10^{-4} in relative modulus over $T = 2$ to 8; the untracked members agree only to 10^{-3} – 10^{-2} , which is expected since the convergence test never monitors them.

Three safeguards complete the iteration in either mode. Every block member is renormalized after each de-mixing, and a member whose norm underflows or overflows is replaced by a fresh random vector. The condition number of S is monitored: a large value means two block directions have effectively merged, and the last member is then replaced by a random vector bi-orthogonalized against the rest with the bilinear form. Convergence is measured on the leading Ritz values (the leading two in production) after matching each

value to its nearest predecessor from the previous iteration, since the frequent modulus-order swaps between near-degenerate eigenvalues would otherwise register as spurious jumps; the iteration stops when the matched movement falls below 10^{-6} , or when it has stopped improving for a set number of iterations.

G.3 Validation against exact diagonalization

At evolution times short enough that the rotated transfer operator stays below a few thousand dimensions, $\mathcal{E}$ can be contracted into a dense matrix and diagonalized exactly. This tests the whole pipeline, builder, rotation, block iteration and pencil, against a reference with no tensor-network approximation in it. Table 4 shows the outcome at $k = 4$ for three operators: the block method reproduces the exact leading pair to between 10^{-8} and 10^{-13} .

operator	dense dimension	exact $ \mu_0 , \mu_1 $	error on the leading pair
ANNNI-type, $VD2$, $p = 0.1$	343	0.9598, 0.3263	1.4×10^{-8}
ANNNI-type, W^{II} , $p = 0.1$	27	0.9758, 0.3292	4.6×10^{-13}
critical Ising	256	1.5488, 0.0611	4.2×10^{-10}

Table 4: **Validation against exact diagonalization.** The block method against dense diagonalization of the same rotated transfer operator, at evolution times short enough for the dense matrix to be tractable. The error is the largest deviation of the two leading Ritz moduli from the exact values.

G.4 Computational cost

One iteration of the block method contains three kinds of tensor-network work. The application step performs $2k$ **tMPO–tMPS** products, each growing the temporal bond from χ to $D\chi$ before compression. The projection step evaluates the k^2 bilinear overlaps of S and M ; no state grows there, and the $k \times k$ dense algebra is negligible. The de-mixing step forms $2k$ linear combinations of k states each. Every combination is built as an exact direct sum, which raises the bond dimension to $k\chi$ before the compression sweep, so both the memory peak of an iteration and most of its work sit in this step. Profiling a production rung ($\chi = 64$, $k = 4$, $p = 0.1$) attributes roughly 85% of the iteration time to the de-mixing sums and their compression sweeps.

Two adjustments reduce this cost without changing what the method converges to. The running direct sum is compressed progressively, each partial sum truncated back to the production bond dimension, with the final addition left exact. And the stagnation patience is shortened, since at late rungs the iteration spends long plateaus below the convergence threshold before the stop triggers. Together the two halve the wall-clock time of the longest rungs while moving $|\mu_0|$ by 4×10^{-6} relative. The mechanisms that reduce the number of iterations, rather than the cost of each one, are described next.

One conceptual point about the truncation belongs here. Each truncation uses only the current states, not the columns of the network still to be absorbed, so a component discarded now could in principle matter later. Translation invariance keeps this harmless: every subsequent column applies the same operator $\mathcal{E}$, which regenerates any part of the leading invariant subspace that has been trimmed away.

G.5 Production options

A thin, model-agnostic builder sits above the solver. It assembles the one-step propagator as an MPO (Appendix F), performs the 90° rotation that maps the spatial virtual bond

onto the temporal physical index, and returns the transfer-matrix MPO with a seed tMPS carrying the right index skeleton. The seed is always overwritten with random complex tensors before the iteration, because a symmetry-special seed converges to a subdominant sector. The temporal physical dimension is read from the operator rather than hard-coded, which is what lets the same code path serve the W^I , W^{II} and $VD2$ kernels and every model without modification.

The options that control a production run are the following; none changes what the method converges to.

Warm starting. Along a ladder of evolution times, the converged block at one time is re-indexed onto the longer time-site set of the next and padded with a small random tail, so the iteration only has to relax the newly added sites. The bond cap and the cutoff follow a schedule that is loose early and tight near convergence: the cap grows in steps of two up to χ , and the cutoff holds at 10^{-8} for the first forty iterations before tightening to 10^{-10} . Warm continuity also stabilises the selection along the ladder, since consecutive rungs then start from the same leading subspace.

Block-size escalation. At stronger frustration the leading $k = 4$ block can fill with members of a single near-degenerate cluster, pushing the subleading eigenvalue needed by Eq. (7) outside the captured window. The driver detects this and escalates the block in steps ($k = 4 \rightarrow 6 \rightarrow 8$), warm-seeding each larger solve with the converged smaller block, so the extra cost is paid only where the spectrum demands it.

Truncation scheme. The truncated object is selectable between **RTM** and **RDM** (Section 3.4), with RTM the production default and RDM the well-conditioned fallback.

Crash safety. Each (p, T) point is written to a per-label cache as soon as it converges, keyed by its parameters, so an interrupted run loses at most the point in flight and a completed point is never recomputed. A separate checkpoint stores the last converged block, so a resubmitted run resumes warm.

G.6 Selecting the dominant eigenvalue

The production rule is the one stated in Section 6.2: the eigenvalue of largest modulus, with continuity against the preceding rung breaking ties below one per cent. Two safeguards protect a ladder built on this rule. A rung whose selected $|\mu_0|$ jumps by more than a prescribed fraction relative to the last trusted value is flagged, since such a jump signals unreliable convergence rather than physics. A flagged rung is not discarded: the physical eigenvalue usually survives further down the block, so a post-processing pass restricts the candidates to a window of five per cent around the last trusted modulus and applies dominance inside it, chaining the reference from the last accepted point. This repairs affected rungs from the stored spectra alone; on the one rung where an independently converged run exists for comparison, the repaired eigenvalue agrees with it to 3×10^{-5} relative. The subleading members are then ordered by their phase distance from μ_0 and matched to the conformal tower rung by rung.

Finally, whenever a limitation is observed, we test whether it is physical or resolution dependent by repeating representative calculations at half the bond dimension, with a tighter truncation cutoff, and with **RDM** rather than **RTM** truncation. A feature is attributed to the transfer matrix only if it persists under all three changes.

G.7 Why eigenvalues outlive eigenvectors

Section 6.2 states that eigenvalues remain accurate where eigenvectors do not. Both halves of that statement follow from first-order perturbation theory for a non-Hermitian

matrix [49]. Perturb $\mathcal{E}$ by $\delta\mathcal{E}$ and expand the perturbed eigenvector as $|R_i\rangle + \sum_{j \neq i} a_{ij} |R_j\rangle$. Projecting the eigenvalue equation onto $\langle L_i|$ gives the shift of the eigenvalue,

$$\delta\mu_i = \frac{\langle L_i | \delta\mathcal{E} | R_i \rangle}{\langle L_i | R_i \rangle}, \quad (49)$$

and projecting it onto $\langle L_j|$ with $j \neq i$ gives the mixing coefficients,

$$a_{ij} = \frac{\langle L_j | \delta\mathcal{E} | R_i \rangle}{(\mu_i - \mu_j) \langle L_j | R_j \rangle}. \quad (50)$$

The two expressions differ in one respect that matters here. Equation (49) contains no eigenvalue difference, so an eigenvalue stays accurate as long as its own overlap $\langle L_i | R_i \rangle$ stays finite. Equation (50) is divided by $\mu_i - \mu_j$, so the eigenvector mixes with its neighbours as soon as two eigenvalues approach, and the de-mixing that separates the block members becomes correspondingly harder.

The size of the denominator in Eq. (49) is measured by the phase rigidity of a biorthogonal pair,

$$r_j = \frac{|\langle L_j | R_j \rangle|}{\|L_j\| \|R_j\|}, \quad (51)$$

the normalized overlap between the left and right eigenvector of the same eigenvalue: it equals one for a Hermitian matrix and falls towards zero as the two rotate towards a common direction. The same expression therefore bounds how far the protection extends, since an eigenvalue tied to an individual eigenvector loses accuracy as its overlap falls. What stays well conditioned is the leading subspace as a whole, whose separation from the rest of the spectrum governs the block iteration. Our eigenvalue-only mode reads the spectrum at that level: it never assigns block members to separate eigenvectors, but carries an orthonormal basis of the subspace and takes the Ritz values from the projected pencil. The eigenvalues therefore behave better than the eigenvectors not because they are immune, but because the subspace-level read avoids the ill-conditioned de-mixing that the eigenvector route cannot do without.

One further diagnostic belongs here. The spectral ratio $|\mu_1/\mu_0|$ controls the suppression of the first subleading contribution in Eq. (9), but it is not a convergence criterion for the block, which is governed instead by the separation of the whole leading subspace from the rest of the spectrum. The two differ near a degeneracy inside the block, where $|\mu_1/\mu_0|$ may approach one while the subspace stays well separated and its eigenvalues keep converging.